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Survey paper

A systematic literature review of emergency communications assisted by unnamed aerial vehicles

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ABSTRACT

Reliable communication systems are critical during emergency scenarios such as floods, earthquakes, fires, and hurricanes to ensure timely information exchange and minimize damage. However, traditional communication infrastructure is often severely compromised during such events. Unmanned Aerial Vehicles (UAVs) have emerged as a promising solution to deploy temporary communication networks in disaster-affected areas. Although several studies have proposed architectures and deployment strategies for UAV-assisted communication, and reviews have addressed their potential, comprehensive discussions on protocol configurations, architectural considerations, and deployment parameter optimization remain limited. This paper presents a systematic literature review covering the period from 2014 to 2024, focusing on the use of UAVs in emergency communication systems. The reviewed articles are classified into a structured taxonomy according to the emergency scenario, UAV role, communication architecture, and protocol layer. Additionally, common communication parameters optimized for deployment, such as coverage and throughput, and the optimization methods employed, including heuristic and artificial intelligence techniques, are identified. The findings reveal that UAVs are predominantly deployed as aerial base stations for large-scale dynamic disasters, with protocol development focusing mainly on the physical, data link, and network layers. However, the lack of standardized deployment guidelines and limited real-world experimentation highlight a significant research gap. This review underscores the need for benchmark methodologies to support the effective and comparable deployment of UAV-assisted emergency communication systems.

1. Introduction

During natural disasters such as earthquakes, hurricanes, and floods, public communication infrastructure is often disrupted, hindering rescue efforts and delaying relief and recovery. Human-made disasters, including explosions, large-scale fires, terrorist attacks, civil unrest, and mass gatherings, can also compromise communication infrastructure through system overload, deliberate sabotage, or physical damage [1,2]. In these circumstances, the rapid restoration of communication systems is essential to enable the timely exchange of critical information on incident assessment, victim status, operational conditions, and situational severity for both civilians and first responders [3,4].

Emergency communication systems (ECS) are designed to support search-and-rescue (SAR) missions, protect lives, and coordinate actions among first responders and civil protection centers. They enhance situational awareness by delivering real-time data that enables informed decision-making [5]. ECS also facilitates the dissemination of safety instructions, evacuation plans, logistics, shelter allocation, and recovery operations. Achieving these functions requires robust and interoperable architectures that integrate sensing and networking elements such as sensors, communication devices, relays, access points (APs), and gateways [6].

Infrastructure-independent technological solutions such as wireless sensor networks (WSNs), mobile ad hoc networks (MANETs), and satellite systems have been explored as alternatives in disaster scenarios. However, large-scale deployment of sensor nodes may be impractical in earthquakes or floods due to high device loss risk, while terrestrial systems often face backhaul bottlenecks [7]. Satellite links operating in the microwave and millimeter-wave bands are also vulnerable to rain attenuation, especially above 10 GHz, requiring careful link budget

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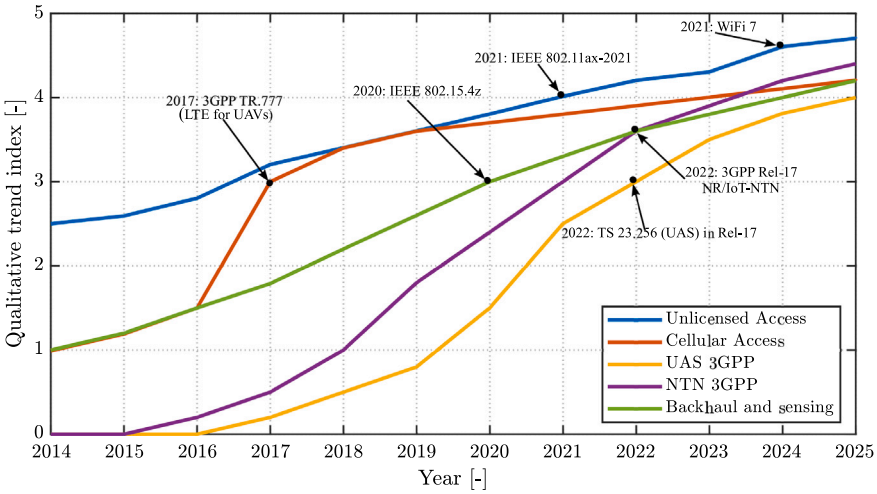


Fig. 1. Evolution of UAV-enabling wireless technologies since 2014. The 0–5 "maturity and attention" index is an ordinal, qualitative trend indicator.

design in severe weather [8]. Mobile terrestrial base stations represent another option, but they frequently suffer from limited backhaul capacity and inefficient deployment [9]. These limitations underscore the need for complementary approaches to ensure reliable ECS under diverse conditions.

In this context, UAVs have emerged as a promising technology to sustain emergency communications. By operating at higher altitudes, they increase the probability of line-of-sight (LoS) links with ground user equipment (UE), extend coverage, and improve connectivity. Their agile repositioning also enables adaptation to user demand and obstacle avoidance, enhancing reliability and capacity during SAR missions [10]. UAVs can thus improve network performance in disrupted environments; however, practical challenges persist, including optimal 3D placement, limited energy and payload, autonomous control, and constrained backhaul capacity [11–13]. Integrated sensing and communications (ISAC) is further expected to expand UAV utility by combining perception and connectivity within a single spectrum and hardware platform [14,15].

Meanwhile, the evolution of wireless technologies has accelerated the integration of UAVs into emergency communication systems. Fig. 1 illustrates how enabling technologies have progressed since 2014, using an ordinal "maturity and attention" index from 0 to 5. Wi-Fi standards evolved from ad hoc access toward dense IEEE802.11ax/11be (Wi-Fi 6/7), while IEEE802.11ah (HaLow) extended low-power range for telemetry and control [16–18]. Cellular support advanced from long term evolution (LTE)-connected aerial UEs to 5G New Radio (5G NR) sidelink for out-of-coverage coordination and UAS identification and tracking [19,20]. The 3rd generation partnership project (3GPP) has also incorporated non-terrestrial features into its standards: Release 17 introduced non-terrestrial networks (NTN) for both NR and internet of things (IoT), later extended with features such as regenerative payloads and reduced-capability devices to enhance satellite-UAV integration [21–23]. Complementary enablers, including mmWave/E-band and free-space optical backhaul, long-range wide area network (LoRaWAN), and ultra-wideband (UWB)-based ranging, have also matured to support resilient, rapidly deployable communications [24–27]. Looking ahead to 6G, the objective is seamless integration of terrestrial and non-terrestrial assets into space-air-ground integrated networks (SAGIN), coordinated through AI-assisted orchestration.

Although recent technological advances have expanded UAV capabilities, persistent challenges still continue to limit their deployment for emergency response. Determining optimal positioning to maximize network coverage during disasters remains complex [28]. Power constraints, including limited battery capacity and the energy demand of onboard communication equipment, directly affect flight duration and overall performance [29,30]. Payload capacity also limits the integration of wireless modules, which are necessary for UAVs to operate as airborne base stations (ABSs) or communication relays [31,32]. These functions are essential to restore connectivity where conventional infrastructure is unavailable, allowing UAVs to complement ground-based networks in diverse disaster scenarios. Current applications can be broadly classified into UAV-assisted ad hoc networks and UAVs integrated with existing cellular infrastructures [33]. Furthermore, issues such as autonomous navigation, control mechanisms, and network capacity must be carefully addressed when designing UAV-based platforms [12,13]. In light of these constraints, it becomes essential to analyze how UAVs have been applied to mitigate such limitations, enhance reliability, and support their integration into emergency response frameworks.

Building on these challenges, it is also relevant to examine how the academic community has approached the role of UAVs in communication systems. General surveys on emergency communication technologies highlight UAVs only briefly, without detailed analysis [3,34–37]. Other works focus on UAV-based communication networks [11,38–41], emphasizing technical performance and network design rather than their specific use in emergencies. A smaller subset of studies explicitly considers UAVs for disaster response [6,42–47], but these analyses remain fragmented. To date, no comprehensive review has systematically examined the types of emergencies in which UAVs have been deployed, nor the communication protocols and network architectures employed. Moreover, the effectiveness of UAV-based systems depends on the careful configuration of parameters such as positioning, coverage, resource allocation, trajectory, and energy management, often addressed through optimization techniques. However, the literature has not yet provided an integrated assessment of how these parameters are optimized to ensure reliable performance in emergency environments.

This paper presents a Systematic Survey (SS) on the use of UAVs to enable communication systems in emergency scenarios, following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology. The review specifically examines the types of emergency scenarios where UAV-based communication systems have been proposed, the communication protocols and architectures applied, and the optimization techniques and parameters employed to enhance system performance. To the best of the authors' knowledge, no previous review has comprehensively summarized these aspects. The main contributions of this study are as follows:

- A classification of emergency scenarios where UAV-based communication systems have been deployed.
- A summary and categorization of communication protocols supporting UAV communications during emergencies.

Table 1
List of abbreviations.

Acronym	Description
3GPP	3rd Generation Partnership Project
5GC	5G Core
ABS/DBS/FBS	Airborne/Drone/Flying Base Station
A2A/A2G	Air-to-Air/Air-to-Ground
AFCA	Adaptive Flocking Control Algorithm
AODV	Ad Hoc On-Demand Distance Vector
AP/MAP	Access Point/Mobile Access Point
AR/AWRN	Aerial Relay/Aerial Wireless Relay Network
BS/GBS	Base Station/Ground Base Station
D2D/D2V	Device-to-Device/Device-to-Vehicle
DTN/NDT	Delay/Disruption-Tolerant Networking/Non-Delay-Tolerant
ECC/ECV	Emergency Communications Center/Vehicle
ECS	Emergency Communications System
EH	Energy Harvesting
EIRP	Equivalent Isotropically Radiated Power
FANET/MANET	Flying/Mobile Ad-Hoc Network
FR	First Responder
FSOC	Free-Space Optical Communications
GA	Genetic Algorithm
GGF	Geographic Greedy Forwarding
HAP/HAPS	High-Altitude Platform/Station
HIL	Hardware-in-the-Loop
IoT	Internet of Things
IRS	Intelligent Reflecting Surface
KPI	Key Performance Indicator
LEO	Low Earth Orbit
LoRaWAN	Long Range Wide Area Network
LPWAN	Low-Power Wide-Area Network
LoS/NLoS/BLoS	Line-of-Sight/Non-LoS/Beyond-LoS
LTE	Long-Term Evolution
MEC	Mobile Edge Computing
mMTC	massive Machine-Type Communications
NFV	Network Function Virtualization
NOMA	Non-Orthogonal Multiple Access
NR	(5G) New Radio
NTN/TN	Non-Terrestrial Network/Terrestrial Network
NVIS	Near-Vertical Incidence Skywave
OCC	Optical Camera Communication
PDR/PER	Packet Delivery Ratio/Packet Error Rate
PL	Path Loss
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
PSO/IPSO	Particle Swarm Optimization/Improved PSO
QoS	Quality of Service
RedCap/eRedCap	(enhanced) Reduced Capability UE
RGB	Red, Green, Blue
RL/DRL/MARL	Reinforcement Learning/Deep RL/Multi-Agent RL
RSSI	Received Signal Strength Indicator
SAGIN	Space–Air–Ground Integrated Network
SAR	Search and Rescue
SDN	Software-Defined Network
SI	Swarm Intelligence
SIB	System Information Block
SLR/SS	Systematic Literature Review/Systematic Survey
SNR/SINR	Signal-to-Noise Ratio/Signal-to-Interference-and-Noise Ratio
SWaP	Size, Weight, and Power
SWIPT	Simultaneous Wireless Information and Power Transfer
TDMA	Time Division Multiple Access
UAV	Unmanned Aerial Vehicle
UE	User Equipment
URLLC	Ultra-Reliable Low-Latency Communications
USN/WSN	Ubiquitous/Wireless Sensor Network
VLC	Visible Light Communication

- An overview and classification of UAV communication architectures applied in emergency contexts.
- A summary of key communication parameters optimized for UAV-based emergency communication systems.
- An overview of optimization techniques employed to enhance UAV communication performance in emergency scenarios.

For clarity, the list of acronyms and abbreviations used throughout this survey is provided in Table 1. The remainder of this paper is organized as illustrated in Fig. 2. Section 2 reviews the most relevant related works on UAV-supported communication systems, particularly those applied in emergencies caused by natural disasters. Section 3 details the methodology adopted in conducting the systematic survey. Section 4 presents the main findings and results of the review, and discusses these results, while Section 5 identifies knowledge gaps and research trends that could guide future research. Finally, Section 6 concludes the paper and outlines potential directions for further investigation.

2. Related works

This section provides an overview of existing surveys related to communication systems in emergency scenarios where UAVs are considered potential communication enablers. The studies are organized into three categories: (i) reviews of emergency communication systems where UAVs are briefly mentioned among other technologies (Table 2); (ii) surveys focused on UAV communication systems without a specific emphasis on emergency applications (Table 3); and (iii) studies specifically addressing UAVs as enablers of communication support for ground users in disaster scenarios (Table 4).

In the first category, Pervez et al. [34] provided a comprehensive review of wireless technologies for emergency response and proposed guidelines for selecting suitable technologies based on scenario requirements. Deepack et al. [35] examined the role of fourth- and fifth-generation cellular networks in post-disaster environments and suggested strategies for spectrum allocation and device-to-device (D2D) communication to improve resilience. Debnath et al. [2] reviewed a wide range of emergency communication standards and future technologies, highlighting the relevance of cognitive radios, MANETs, and cloud-based systems. Carreras-Coch et al. [37] investigated communication technologies deployed in disaster environments and proposed a hybrid architecture for mobile agents under harsh conditions, integrating UAVs, Near-vertical incidense skywave (NVIS), and IoT-based systems. Wang et al. [3] offered a broad survey of emergency communication technologies, including satellite, ad hoc, and private wireless networks, and introduced the concept of integrated broadband and narrowband networks.

A summary of the key findings from these studies is presented in Table 2. Although these reviews provide valuable insights into communication technologies for emergency response, they do not specifically address UAV-based systems or the unique challenges associated with their deployment.

The second category includes surveys focused on UAV communication systems with no specific emphasis on emergency scenarios. Zeng et al. [11] examined the integration of UAVs into 5G and future cellular networks, organizing their work into two frameworks: Wireless communications supported by UAVs and cellular-connected UAVs. Their study highlighted key challenges and considerations for these applications.

Sharma et al. [48] reviewed UAV communication technologies across centralized, decentralized, and hybrid architectures. They addressed routing protocols for single UAVs and swarms, with an emphasis on integration with WSNs, IoT, AI-driven optimization, and navigation strategies to improve efficiency and security. Hentati and Fourati [33] focused on multi-UAV systems, testbeds, and simulation environments, addressing Quality of Service (QoS) requirements and security concerns, while reviewing protocols for high-mobility aerial networks and cross-layer optimization approaches.

Chen et al. [49] systematically analyzed UAV swarm communications, covering star, mesh, and hierarchical architectures, and assessed their advantages, limitations, and application scenarios. They also reviewed routing protocols such as time division multiple acess (TDMA) and non-orthogonal multiple access (NOMA), identifying open challenges in coordination, latency, and security. Similarly, Alsuhli et al. [50] proposed a classification of UAV communication systems

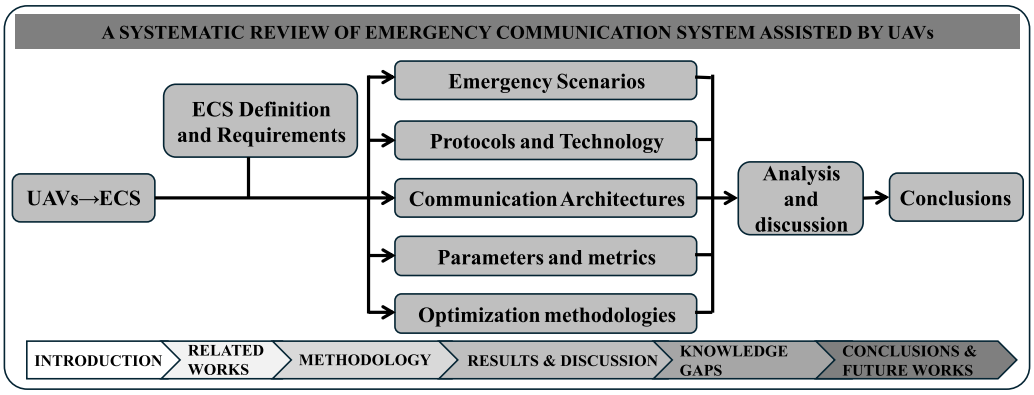


Fig. 2. Organization of the systematic survey.

Table 2
Surveys about emergency communications systems.

Author year	Title	Scenario case of study	Communication technology	Problem and methodology	Results
Pervez et al. [34]	Wireless technology for emergency response: A comprehensive review and some guidelines	Natural disasters, terrorism and battlefield scenarios, and patient monitoring.	Wireless Technologies Development (W4D) enabling Global Access to the Internet All (GAIA)	Compare wireless technologies, to choose the right technology and System according to the different emergency scenarios. Details challenges such as interoperability, bandwidth, and security.	Architectures of wireless-based emergency response systems
Deepack et al. [35]	An overview of post-disaster ECSs in the future networks.	Post-disaster scenarios where existing infrastructure is congested, partially functional, or isolated.	LTE and 5G, D2D communication, IoT, MANET, and satellite.	The telecommunication infrastructure should be embedded with auto-configuration and context-aware network algorithms. Proposes spectrum allocation frameworks and resource management strategies.	Framework, D2D communication, drone-assisted communication MANET, and IoT for Post-disaster scenarios
Debnath et al. [2]	A comprehensive survey of emergency communication network and management.	Pre, during, and post-disaster communication, compatibility with legacy networks, and emergency alerts.	MANETs, LTE, 5G, cognitive radios, and satellite, ABS, D2D, and cloud networks.	Suggest the use of lightweight transceivers. ABSs, and machine-learning (ML) algorithms to optimize performance (resilience and reliability), and reduce congestion. Discusses protocols for spectrum sharing, network scalability, and energy efficiency.	Highlights challenges and solutions to enhance Emergency Communication networks performance.
Carreras et al. [37]	Communication technology in emergency situations	Communications in emergencies.	Ubiquitous Sensor Networks USN.	Analyze surveys on technologies for emergency communications systems and technologies. Combines IoT, UAVs, NVIS, and edge computing.	Reviews EC technologies, and proposes a hybrid architecture.
Wang et al. [3]	An overview of the emergency communication network	A temporary emergency communication network.	Satellite, Ad Hoc, cellular, and wireless private networks.	Proposes a broad band-narrow band integrated emergency communication network.	Identifies challenges and future directions like SAGIN.

based on functional roles (base stations, relays, and end-users) and surveyed enabling technologies ranging from LTE, 5G, and satellite networks to millimeter-wave, low-lower wide-area networks (LPWAN), and free-space optical (FSO) Communications.

Other studies explored advanced frameworks and future directions. Ray et al. [51] investigated UAV-as-a-Service within the 6G SAGIN, addressing key enablers and architectural constraints. Lu et al. [38] and Beegum et al. [39] examined routing algorithms for flying ad hoc networks (FANETs) to achieve seamless coverage in SAGINs, with attention to bio-inspired and adaptive protocols. Zhou et al. [40] surveyed aerial integrated networks for 6G, covering UAVs, high-altitude platforms (HAP), and low-earth orbit (LEO) satellites. Kurunathan et al. [41] reviewed ML applications in UAV communications, identifying perception, planning, feature extraction, and control as key components, and proposed an end-to-end framework for UAV autonomy. Finally, Wan et al. [52] analyzed UAV-based network innovations, emphasizing challenges such as spectrum allocation, autonomy, and energy limitations, and discussed the role of ML and game theory in optimizing connectivity and performance.

Table 3 summarizes the methodologies, scenarios, and contributions of these surveys. Although these works significantly advance understanding of UAV communication technologies, they do not explicitly address emergency response applications or the specific constraints and optimizations required in disaster scenarios.

The third category provides an overview of the most relevant surveys on the application of UAVs for emergency communication systems, focusing on operational scenarios and the communication technologies employed. Erdelj and Natalizio [42] identified key challenges in UAV-based disaster management, including the use of aerial relay networks for wireless sensor systems.

Zhao et al. [43] proposed a unified framework of UAVs as Flying Base Stations for two scenarios: cooperating with surviving terrestrial stations after a disaster and autonomously extending coverage when

Table 3
Surveys about UAVs communications systems.

Author year	Title	Scenario case of study	Communication technology	Problem and methodology	Results
Zeng et al. [11]	Accessing from the Sky: A tutorial on UAV communications for 5G and beyond.	UAV uses include aerial inspection, search and rescue, and communications.	5G	Advances in UAV-assisted wireless and cellular communications addressing 3D mobility challenges.	UAV-assisted wireless communications, and cellular connected UAVs
Sharma et al. [48]	Communication and networking technologies for UAVs: A survey.	Multi-drone, cellular, WSN, and disaster management networks.	WiFi, Bluetooth, ZigBee, Acoustic communication, cellular technology.	Optimization to save power and reduce latency in UAV communications, propose computational offloading to alleviate onboard resources and TDMA for BLoS.	Current state of UAV communication technologies, advances, and challenges.
Hentati et al. [33]	Comprehensive survey of UAVs communication networks.	UAV swarms applications, routing, and energy-efficient protocols.	UAV swarm communication. Routing technologies	ZRP and TORA hybrid routing protocols for UAV swarm control and coordination.	Review UAV swarm architectures and routing protocols.
Chen et al. [49]	Review of UAV swarm communication architectures and routing protocols.	Multi-UAVs for cooperative task, and delay tolerant networks.	Star, mesh and hierarchical topologies, cloud computing, TDMA and NOMA.	Multi-UAV coordination, dynamic routing protocols and security issues.	Reviews architecture challenges and multi-UAV communication design
Alsuhli et al. [50]	A survey on the role of UAVs in the communication process: A technological perspective.	UAVs as relays, and multi-hop network operation.	WiFi, LTE, and mmWave networks, FSO, LPWAN technologies.	Data collection optimization with TSP, and maintaining LoS propagation to enhance Ground-to-UAV link performance and reliability.	Discusses UAV communication roles technological advances.
Ray et al. [51]	A review on 6G for space-air-ground integrated network: Key enablers, open challenges, and future direction.	6G-SAGIN. Singular and multiple UAV-as-a-Service (UAVaaS)	Satellite, mobile technology, mobile backhaul designs and the use of mmWave and microwave.	Adresses challenges like agility and integration in SAGIN, emphasizing advanced communication technology.	UAVs and satellite roles in SAGIN, and their impact on communication agility.
Lu et al. [38]	UAV Ad Hoc network routing algorithms in space-air-ground integrated networks: Challenges and directions.	FANETs that make up a SAGIN.	5G and 6G.	Explores seamless all-area, all-time coverage, which requires efficient routing algorithms.	Reviews state-of-the-art routing for SAGIN integration.
Beegum et al. [39]	Optimized routing of UAVs using bio-inspired algorithm in FANET: A systematic review.	FANET routing protocols with bio-inspired techniques.	UAVs relay nodes, and Cognitive Radios (CR).	Compares non-hybrid and hybrid bio-inspired algorithms, detailing metrics and tools.	Review of bio-inspired routing strategies used in FANET.
Zhou et al. [40]	Aerospace integrated networks innovation for empowering 6G: A survey and future challenges.	AINs for 6G. with UAVs, HAPs, and satellites	6G.	Reviews state of the art methodologies and models in AIN for 6G.	Surveys AINs potential enhancing 6G capabilities.
Kurunathan et al. [41]	Machine learning-aided operations and communications of UAVs: A contemporary survey.	ML applications in UAV communications.	ML.	Analyzes ML-based solutions for IoT, smart cities and UAV.aided edge computing	Proposes an End to end ML framework for UAV operations.
Wan et al. [52]	Advancements and challenges in UAV-based communication networks: A comprehensive scholarly analysis.	UAV use in Disaster, agriculture, security, and rural connectivity.	Beamforming, FSOC, cognitive radio, and RL.	Examines challenges like Battery life, spectrum utilization, autonomy, and system integration.	Offers insights into UAV-based communication innovations and future directions.

no base stations remain, leveraging multihop communication and trajectory optimization. Okada et al. [44] reviewed aerial wireless relay networks (AWRN) where UAVs operate as AP, analyzing the effects of UAV inclination and mobility on signal quality and packet delay.

Yao et al. [45] proposed a Multi-Tier Heterogeneous UAV Network-based Emergency Wireless Communications Network as a temporary architecture to meet urgent communication needs in harsh environments. Alam et al. [46] surveyed Q-learning-based routing protocols for position-aware communication in FANETs, providing insights into performance metrics such as delay, robustness, and signal-to-interference and noise ratio (SINR). Carvajal et al. [47] reviewed challenges and solutions for post-disaster UAV-enabled systems, focusing on path planning and energy optimization through genetic algorithms and ML. Shah et al. [6] proposed a UAV-based emergency communication system architecture using FANETs compatible with existing infrastructure, employing cluster-based topology control to optimize throughput and latency. Chandran and Vipin [28] examined multi-UAV deployments for disaster monitoring in indoor and outdoor environments, emphasizing the integration of edge computing, artificial intelligence, and advanced routing protocols to enhance responsiveness and network resilience.

Despite these advances, current literature lacks comprehensive analyses integrating network architectures, formation strategies, communication protocols, and advanced technologies such as edge computing and artificial intelligence to optimize network performance and enhance security in multi-UAV systems for emergency scenarios. Table 4 summarizes these studies, detailing scenarios or case studies, communication technologies used, addressed problems, methodologies, and obtained results.

Although research on UAV-based communication systems continues to grow, systematic analyses specifically devoted to emergency scenarios remain scarce. Existing surveys often adopt broader approaches, covering applications in agriculture, surveillance, and logistics, while

Table 4
Surveys about UAV-assisted emergency communications system for ground users.

Author year	Title	Scenario case of study	Communication technology	Problem and methodology	Results
Erdelj et al. [42]	UAV-assisted disaster management: Applications and open issues.	UAVs in disaster management with WSN and cellular	Aerial relay network	UAVs for monitoring, logistics, damage assessment, search and rescue.	Classification of the WSN and UAV-assisted disaster applications.
Zhao et al. [43]	UAV-assisted emergency networks in disasters.	UAV cooperation with surviving BSs or as standalone.	Flying base stations	Trajectory optimization, scheduling, multi-hop D2D relaying.	Unified framework for a UAV-assisted emergency networks.
Okada et al. [44]	An overview of aerial wireless relay networks for emergency communications during large-scale disasters.	AWRNs use UAVs as APs, forwarding data directly or via mobility when out of range.	AWRN for DTN	Effects of UAV inclination on communication range and delay across three mobility models: RWP, Rebounding, and CFP.	Simulation results showed there is an Optimum moving.
Yao et al. [45]	The rise of UAV fleet technologies for emergency wireless communications in harsh environments.	Multi-tier UAV architecture for emergency response.	5G and 6G	Resource allocation, full coverage, energy efficiency.	Hybrid-channel resource allocation, RIS-assisted UAV communications.
Alam et al. [46]	Survey on Q-Learning based position aware routing protocols in flying Ad Hoc networks.	FANETs with terrestrial/non-terrestrial sections.	BS	Q-Learning for position-aware routing, analyzing SINR, delay, robustness.	Insights into routing optimization and self-healing mechanisms.
Carvajal et al. [47]	An overview of UAV-enabled communications system in post-disaster emergency scenarios.	Maintain communication services after damage to the communications network caused by disasters.	5G WiFi WiMAX	Path planning and energy optimization using ML and GAs.	Overview of the challenges and solutions for the feasibility of ECS in different scenarios.
Shah et al. [6]	Architecture of emergency communications systems in disasters through UAVs in 5G and beyond.	FANETs, able to communicate with existing networking infrastructure.	5G and 6G	Cluster-based topology control to optimize throughput, delay, and outage probability.	Simulation results (message delay)
Chandran et al. [28]	Multi-UAV networks for disaster monitoring: challenges and opportunities from a network perspective	Multi-UAV systems for indoor/outdoor disaster response.	Cellular (2G, 3G, 4G), WiFi, WiMAX, Bluetooth, and ZigBee.	Routing protocols, task offloading, ad hoc networking, edge computing, AI.	Identifies challenges in deployment, data processing, and network architecture.

overlooking the operational and technical requirements of disaster response. For instance, Hentati et al. [33] and Alsuhli et al. [50] provide overviews of UAV communication architectures but do not explicitly focus on emergency management. Consequently, there is a lack of rigorous, methodologically structured surveys addressing UAV-supported emergency communication systems.

Furthermore, current taxonomies of UAV communication architectures often omit the specific functional requirements of emergency applications. Classifications such as those by Chen et al. [49] emphasize network topologies (e.g., ring, star, mesh) and broader categories (e.g., VANETs, MANETs), but they do not explicitly address disaster scenarios. Detailed classifications of communication protocols according to the Open Systems Interconnection (OSI) model, outlining implementations at the physical, media access control (MAC), network, and application layers and evaluating their suitability for emergency operations, are largely missing.

Additionally, cross-sectional analyses linking optimization parameters to performance metrics are absent. Key indicators such as latency, throughput, coverage, energy efficiency, and mobility have not been systematically analyzed, nor have corresponding optimization strategies been applied. Existing reviews, including Beegum et al. [39], focus mainly on FANET optimization using bio-inspired algorithms. Therefore, there is an opportunity to comparatively analyze optimization methods for these metrics. Furthermore, research gaps remain in the representation of application-layer protocols, the standardization of performance parameters, the integration of heterogeneous communication technologies, and the development of experimental testbeds to validate real-world applicability.

3. Methodology

Systematic surveys play a crucial role in informed decision-making and should be conducted with transparency and completeness to allow replication or future updates. To achieve this, we adhere to the PRISMA guidelines [53]. This approach consists of four key stages: (1) defining research questions, (2) establishing a search strategy, (3) selecting relevant studies, and (4) analyzing and synthesizing the information gathered.

3.1. Research questions

Systematic surveys play a crucial role in informed decision-making and must be conducted with transparency and completeness to enable replication and future updates. To ensure methodological rigor, this study follows the PRISMA guidelines [53], which consist of four key stages: (1) defining research questions, (2) establishing a search strategy, (3) selecting relevant studies, and (4) analyzing and synthesizing the information gathered.

3.2. Research questions

A systematic review begins with the definition of research questions (RQs) to guide the search and selection of studies and to frame the analysis of UAV-supported ECS, particularly in the context of natural and human-made disasters. The extraction of relevant data from the literature is also structured around these questions. The RQs defined for this survey are the following:

- RQ1: How can UAVs be employed to support emergency communication systems?
- RQ2: What communication protocols are utilized in UAV-based emergency communication networks?
- RQ3: What UAV-based communication architectures are implemented during emergency scenarios?
- RQ4: Which UAV communication parameters can be modified and optimized in emergency contexts?
- RQ5: What optimization techniques and methodologies are applied to enhance UAV communication performance during emergencies?

These RQs were designed to capture the state of the art in UAV-enabled emergency communications. RQ1 investigates the scenarios and use cases where UAVs have been deployed for disaster response. RQ2 and RQ3 address the enabling technologies of UAV-assisted communications: RQ2 focuses on the communication protocols employed, whereas RQ3 examines the architectural models proposed. Given that UAV placement and configuration are critical for maintaining network coverage, RQ4 identifies the communication parameters that must be adapted during emergencies, such as transmission power, frequency, and altitude. Finally, RQ5 explores the optimization methods and algorithms used to determine these parameters effectively.

3.3. Search strategy

The systematic survey was conducted to address the proposed research questions, following a structured search in two leading scientific databases: Web of Science (Clarivate) and Scopus (Elsevier). These sources were selected for their broad coverage of peer-reviewed publications in disciplines relevant to this study, including computer science, engineering, and communications. Using both databases reduces potential biases associated with indexing policies or reporting standards, ensuring that data are retrieved from reputable and widely recognized outlets.

The search targeted peer-reviewed journal articles and review papers, with preference given to highly cited and foundational contributions most relevant to the research objectives. Keywords were carefully designed to capture the intersection of ‘‘UAV communications’’ and ‘‘emergency communications’’ within the context of disaster scenarios. To ensure precision, the search was limited to article titles and abstracts, which typically provide concise summaries of each publication’s scope and relevance.

3.4. Screening process

Based on the search query illustrated in Fig. 3, a total of 102 records were initially retrieved from the selected databases: 78 from Scopus and 24 from WoS. Among these, 21 WoS records were duplicates of those already identified in Scopus and were removed, leaving 81 unique records for screening. The screening process was then conducted in two stages to ensure alignment with the research objectives and exclusion of irrelevant or ineligible publications:

1. **Title and abstract screening:** Titles and abstracts were systematically reviewed to assess relevance. As a result, 23 publications were excluded for being outside the scope of this review.
2. **Full-text examination:** The remaining articles were examined in detail, with particular attention to the introduction, stated contributions, results, and conclusions. This stage led to the exclusion of 11 additional records that did not meet the analysis criteria.

Following this two-stage evaluation, 47 articles were retained to form the basis of the final analysis.

3.5. Inclusion and exclusion criteria

The inclusion and exclusion criteria were defined to ensure methodological rigor and maintain alignment with the scope of UAV-assisted ECSs (see Fig. 4).

Studies were included if they: (i) directly addressed one or more of the research questions; (ii) were published as peer-reviewed journal articles or review papers; (iii) appeared between 2014 and 2024; and (iv) were written in English or Spanish.

Conversely, studies were excluded if they met any of the following conditions: (i) duplicate entries; (ii) non-peer-reviewed publications, including preprints and white papers; (iii) thematic misalignment, particularly when the primary focus did not involve the design, optimization, or evaluation of UAV-based ECSs; or (iv) inaccessibility through an online electronic database.

3.6. Synthesis

The selected studies were systematically analyzed following the screening and selection process to extract insights relevant to the five RQs. Each article was classified into one or more thematic categories according to its alignment with RQ1–RQ5. The analysis combined quantitative and qualitative perspectives: the quantitative assessment determined the frequency and distribution of research focus areas, while the qualitative analysis identified recurring themes, gaps, and methodological patterns across the literature.

The results were processed and visualized using MATLAB and Microsoft Excel. MATLAB was employed for data processing and plotting, whereas Excel supported structured data organization and preliminary trend identification. This combination facilitated the generation of informative visualizations that emphasized patterns, correlations, and research distributions across the reviewed studies. The methodological framework thus integrated qualitative insights, quantitative assessment, and comparative analysis, providing the foundation for the comprehensive results discussed in the next section.

4. Results and discussions

This section presents the key findings for each RQ from the analysis of the selected studies. For interpretability, the RQs are grouped into two thematic areas: (A) scenarios, use cases, and the associated UAV communication functions; and (B) optimization of communication parameters for UAV communications in emergencies. Furthermore, this section presents a structured discussion of the findings derived from a systematic review of 47 peer-reviewed articles. These studies were selected according to defined inclusion criteria and organized by thematic focus, methodological approach, and technological contribution.

4.1. Scenarios and UAV communication functions

4.1.1. Research Question 1

To address RQ1 (*How can UAVs be employed to support emergency communication systems?*), we analyzed deployment scenarios subject to environmental, technical, and geographical constraints that impose specific requirements on UAVs equipped with communication payloads for emergency connectivity.

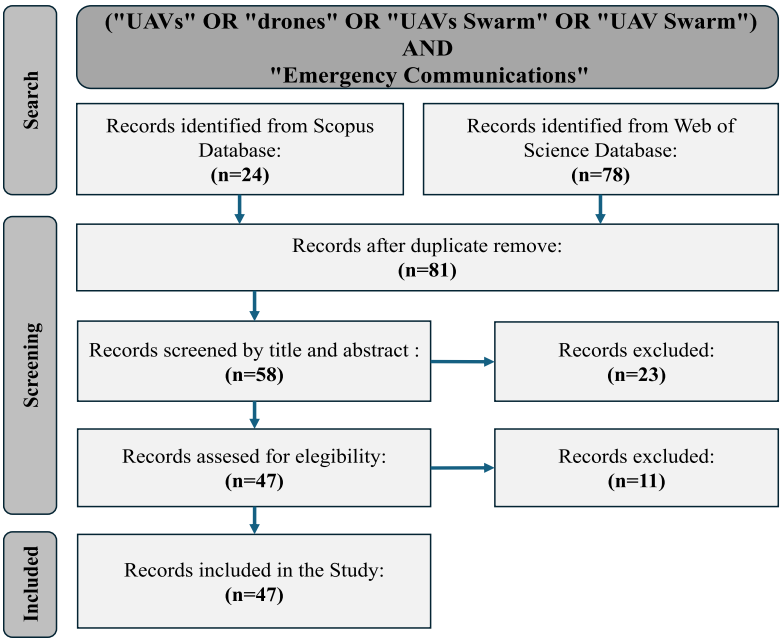


Fig. 3. Illustration of the search and screening process followed to perform the systematic literature review.

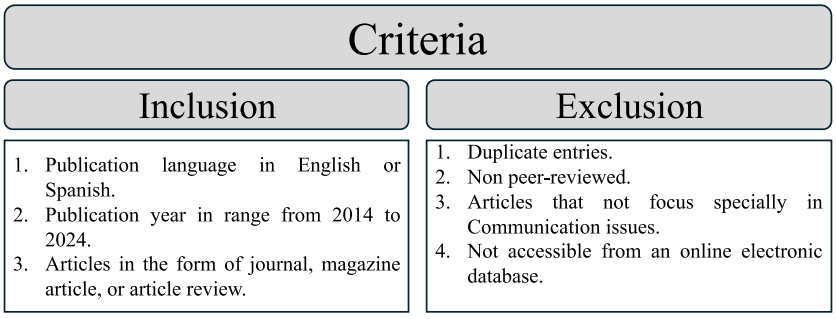


Fig. 4. Inclusion-exclusion criteria.

Case studies reported in the literature document applications in natural and human-made disasters, detailing challenges, implementation strategies, applications, and observed results. Based on convergent evidence, articles were assigned based on their central contribution or functionality in the emergency communication context, to the following categories:

- **UAV Coverage Restoration:** This category encompasses systems where one or more UAVs operate as aerial base stations, providing direct wireless access in the affected area, effectively acting as temporary cellular sites or Wi-Fi hotspots. The UAV-to-user link is primary; backhaul is presumed via existing infrastructure or a simple point-to-point link. Inter-UAV multi-hop is typically absent, with multiple UAVs covering sectors independently. Studies focus on coverage footprint, altitude/placement, and capacity when ground infrastructure is unavailable. The defining objective, regardless of the type of disaster, is rapid restoration or extension of coverage.
- **UAV Relay and Multi-Hop Networks:** This category comprises systems where UAVs act as communication relays or routers, establishing one or more hops between ground nodes or base stations. Data is relayed across UAV nodes to overcome distance or obstruction, forming an aerial backhaul or mesh. Rather than broad end-user access, the goal is to reconnect isolated segments via store-and-forward or real-time routing, with at least one UAV serving as an intermediate node. Key focus include multi-hop algorithms, relay placement, and routing design. This spans single-UAV one-hop bridges and multi-UAV ad hoc networks in any disaster that creates a communication gap.
- **Edge Computing UAVs for SAR:** This category is defined by onboard processing and mission-specific data services on UAVs during emergencies. Drones act as edge nodes, performing real-time analytics to support SAR. We include work where UAV analysis produces actionable outputs, such as survivor location or damage assessment, thereby reducing latency and dependence on remote computing. Communication remains essential to deliver processed data to ground teams or the cloud, but the novelty lies in leveraging onboard intelligence. Any study treating the UAV as a flying edge server or tightly integrating with ISAC, IoT, wearables falls here. Unlike first and second categories, coverage and multi-hop are secondary; data timeliness and local analytics are the decisive factors. In mixed-role studies, classify here when computing for SAR is the central emphasis.
- **Autonomous Swarm Networks:** This category covers multi-UAV systems with autonomous coordination to provide emergency communications across wide or rapidly changing areas. A networked drone team adapts positions, roles, and links self-organizing via swarm control, dynamic deployment, or distributed resource allocation. Studies emphasize decentralized decision-making and real-time repositioning to sustain coverage and link quality as conditions evolve. Although often applied to

Table 5
UAV-based emergency communication scenarios: categories, references, and share of studies.

Category	References	%
UAV coverage restoration	[6,30,37,54–64]	29.8
UAV relay and multi-hop networks	[7,9,29,43,44,55,57,65–71]	29.8
Edge computing UAVs for SAR	[72–77]	12.8
Autonomous swarm networks	[12,13,46,78–89]	31.9

large-scale or highly dynamic disasters, classification depends on architecture; any self-adjusting multi-UAV network qualifies.

Assignments are not mutually exclusive: Studies can fall into multiple categories when appropriate, each study was assigned to the category that best represents its primary contribution. Table 5 summarizes the classification, considering scenario-specific operational requirements, technical capabilities of communication systems, scope of application, and potential end users.

Nearly one third of the studies examine deployments in natural disasters where infrastructure is severely damaged. UAVs operate as aerial base stations to re-establish connectivity, often leveraging small-cell LTE/5G or Wi-Fi, and are optimized for placement, energy use, and load balancing to sustain service over time [30,49,54,63,65,78,87]. Although cellular systems offer standardized, interoperable coverage and high throughput suited to time-critical coordination, costs can be substantial when satellite backhaul or dedicated ground support are required.

An equal fraction of studies addresses UAV-assisted relay and multi-hop networking, reconnecting isolated segments when direct links are obstructed. D2D and ad hoc multi-hop schemes are common; Wi-Fi is widely used for short-range, rapid deployment (e.g., pop-up hotspots during the initial response) [9,57,75,79,81,86]. Hybrid designs that combine, for example, Wi-Fi with LTE or cognitive radio with FSO, improve adaptability across terrains, user densities, and infrastructure conditions.

A smaller but growing body of work considers mobile edge computing (MEC)-enabled UAVs that process data at the edge to lower latency in real-time tasks (e.g., live video, distributed sensing, autonomous navigation). These platforms have demonstrated fast image analysis for victim detection, telemedicine support, and rapid sensor fusion in time-pressured settings [74,75,77].

Finally, the largest group explores autonomous swarms for adaptive coverage, dynamic repositioning, and resilient networking. Distributed control (clustering, reinforcement learning, decentralized decision-making) underpins scalability and robustness without centralized coordination. Robotics-inspired models (e.g., flocking/Boids, Voronoi partitioning) support geographic task allocation and stable connectivity [86,89]. Trajectories are either predefined (circular, grid search, hovering) or updated online through AI-driven policies.

The above classification clarifies how UAVs are employed across emergency scenarios: restoring coverage, bridging segments via relays and multihop, enabling SAR with edge processing, and coordinating as autonomous swarms. To understand what makes these deployments viable in practice, we next examine the underlying communication technologies that support each role. This technology-centric view complements the scenario taxonomy by linking required functions (e.g., rapid access, resilient backhaul, delay-sensitive sensing) to specific radio interfaces and architectural choices (LTE/5G NR, NTNs, LPWANs, mmWave/FSO, terahertz, ISAC, and AI-enabled control). Analyzing in terms of operational maturity versus research attention provides a consistent basis to situate each scenario class within a realistic stack of enabling technologies and highlights where integration across space-air-ground layers can strengthen robustness during disaster response.

Over the past decade, LTE has served as a dependable baseline for aerial base stations and aerial user equipment, enabling rapid service restoration through optimized 3D placement, interference control, and load balancing. Repeated field-style results support its high maturity for deployable links in emergencies [6,54,76,85,88].

Terrestrial 5G NR increases throughput and reduces latency for video, telemetry, and command-and-control traffic. In practice, NR-based systems are commonly integrated with learning-based association, joint resource-trajectory optimization, and clustering. Although altitude-dependent interference remains an active research topic, NR-enabled pop-up coverage is generally considered technically mature for time-critical missions and continues to undergo field evaluation [6,7,56,59,78,87,89].

When backhaul is impaired, NTN networks, including LEO satellites and HAP Systems, extend reach through UAV gateways. Core capabilities are mature, while seamless orchestration, cross-layer resource control, and link selection across space, air, and ground layers remain open challenges [6,37,69,70,78].

LPWAN Technologies such as LoRaWAN, Narrowband IoT, and LTE for machines (LTE-M) offer energy-efficient telemetry and delay-tolerant messaging that complement broadband layers. They suit wide-area sensing, status beacons, and low-rate control and integrate well with edge analytics, indicating medium to high operational maturity [37,73,76].

mmWave and FSO provide multi-gigabit air-ground backhaul. Work on placement, beam steering, hybrid radio frequency (RF)/FSO switching, and weather-aware control shows feasibility. Alignment, blockage, and size-weight-power limits still constrain broad use, so maturity is medium with strong recent attention [9,61,69,70,83].

THz bands target very short-range LoS offload in dense scenes. Joint trajectory and resource designs are promising, but propagation loss, blockage, and hardware readiness remain hurdles. THz is emerging with low to medium maturity [82,83].

ISAC on UAVs aims to unify sensing and data links for search and rescue, mapping, and collision avoidance. Prototypes increasingly pair ISAC with on-board or edge inference, but standardized metrics and large-scale validation are still developing. Attention is rising with moderate maturity [74,76,77].

AI-Enabled Control, such as, RL, multi-agent variants, and meta-heuristics coordinate placement, routing, spectrum access, and association under non-stationary conditions. Studies report gains in latency, coverage, and energy, mostly in simulations or pilots. Robustness and certification for safety-critical use remain open [7,59,87–89].

FANETs and SAGINs combine these radios and controls into resilient multi-hop and multi-layer systems. Benefits in continuity and redundancy are clear, but unified orchestration and standard interfaces are still needed, placing maturity at medium with sustained momentum [6,12,37,57,71,78,81].

4.1.2. Research Question 1 analysis

The evidence base is heterogeneous and unevenly validated. While autonomous, swarm-based strategies are prominent, many studies presuppose degrees of coordination, localization, and collision avoidance that are not yet routine in field operations. Conversely, on-board edge computing for time-critical SAR remains underrepresented relative to its relevance for latency-sensitive perception and decision tasks. Overall, large-scale trials in realistic post-disaster settings are scarce, revealing a persistent gap between simulation-driven results and deployable practice.

From a technological standpoint, LTE and terrestrial 5G NR serve as high-maturity anchors for near-term UAV links, providing rapidly deployable access and backhaul with well-characterized performance envelopes. NTN and SAGIN offer medium maturity extensions in reach and resilience, whereas LPWAN technologies support low rate sensing and telemetry. In contrast, emerging capabilities, such as hybrid mmWave and FSO backhaul, THz short-range links, ISAC, and swarms of multiple UAVs orchestrated by artificial intelligence, continue to attract substantial research attention but require broader field evidence and harmonized control and interface standards before routine deployment in emergency scenarios [57,59,70,76].

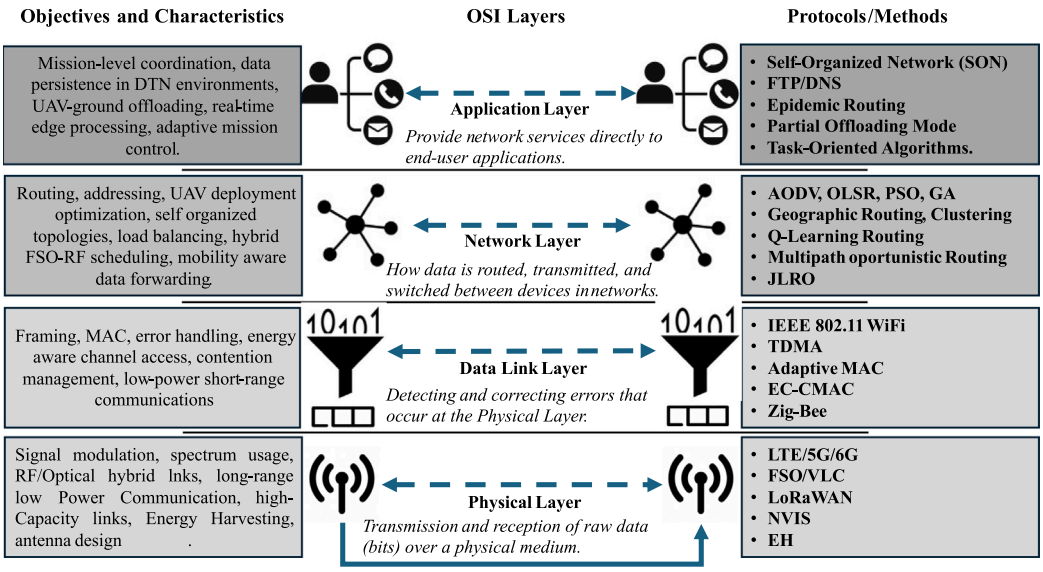


Fig. 5. Observed protocol stack in UAV emergency networks: mapping to OSI layers and indicative data flow.

4.1.3. Research Question 2

RQ2 (*What communication protocols are utilized in UAV-based emergency communications networks?*) was addressed by systematically extracting the protocols reported in the selected studies restricted to emergency deployments. Here, *communication protocols* denote standardized rules, formats, and procedures, covering syntax and packet formats, error control, flow or congestion control, and session management, that govern how data are transmitted, received, and interpreted between network nodes. Protocols operate at different layers of the OSI model. Techniques or heuristics without a formal specification were excluded.

The analysis concentrates on the first three OSI layers, which directly address core challenges in UAV-assisted emergency response, dynamic topologies, intermittent connectivity, time-critical traffic, bandwidth limitations, and constrained processing when infrastructure is degraded or unavailable [90,91]. The Physical layer provides the basis for wireless transmission [92]; the Data Link layer enables reliable node-to-node communication via framing, error detection, and medium access [93]; and the Network layer supports routing and topology management across heterogeneous segments [94,95].

About one sixth of the studies address Physical-layer technologies (LTE/5G radio access, FSO/VLC optical links), reflecting the importance of link feasibility and capacity under degraded infrastructure. Data Link contributions propose MAC strategies tailored to mobility and congestion (e.g., TDMA variants, energy-aware EC-CMAC), whereas Network-layer works emphasize routing under intermittent connectivity, with ad hoc on-demand distance vector (AODV), and delay-tolerant networking (DTN), as well as learning-based route selection. Cross-layer coupling is common in practice (e.g., trajectory or power-control decisions that affect MAC contention and route stability) but remains weakly standardized across the corpus.

Once the communications protocols used in each paper were identified, they were classified according to the layers of the OSI communication model. The referred protocols functions, advantages, and disadvantages are presented in Table 6 focusing primarily on the Physical, Data Link, and Network layers. Besides, Table 7 presents a summary of the OSI layers percentage distribution. Fig. 5 shows the data flow over the different OSI model communication layers and shows the protocols used in each layer.

About 17% of the studies address physical-layer technologies enabling UAV-to-UAV or UAV-to-ground communication. Common choices include LTE/5G, Wi-Fi, mmWave, FSO, and LoRa-WAN. LTE remains dominant due to its maturity and integration with existing infrastructure [54], while FSO and Visible Light Communication (VLC) provide high-capacity optical links but require LoS and are sensitive to weather [9,64]. LoRaWAN supports long-range, low-bandwidth transmissions with minimal power consumption, ideal for isolated, sensor-rich deployments [69]. Advanced technologies are also emerging: NOMA enhances spectral efficiency via power-domain multiplexing [65], energy harvesting (EH) enables energy autonomy, and Terahertz communication promises ultra-high data rates [82]. Techniques like NVIS and conformal antennas extend coverage and integration capabilities [37,62]. Each solution targets specific trade-offs among energy efficiency, interference resistance, and deployment cost.

Approximately 10.6% of the reviewed studies focus on MAC-layer innovations tailored to UAV mobility, intermittent links, and dynamic topology. Standard protocols like IEEE 802.11 (CSMA/CA) and TDMA are widely used but face limitations under high mobility and congestion. Adaptive MAC protocols adjust parameters like contention windows and duty cycles in response to traffic and channel conditions, ensuring energy efficiency and collision mitigation. Priority-based schemes ensure low latency for critical traffic [60]. EC-CMAC protocols integrate cognitive spectrum access with energy-efficient scheduling, enabling opportunistic spectrum use under crowded conditions [60,71]. While standardized MAC protocols offer reliability and interoperability, customized MAC solutions better address the agility and responsiveness demanded in emergency scenarios, particularly in UAV swarms and FANETs.

Network layer solutions are among the most common (12.8%), focusing on routing, topology management, and resource allocation. Routing protocols must operate in real-time, adapt to topology changes, and maintain efficiency under intermittent connectivity. Geographic routing leverages UAV positional data to reduce control overhead, while Q-learning and mobility prediction improve link stability [12,96]. Clustering techniques, such as D2D-based and K-means-based approaches [55,56], reduce UAV relay loads and increase scalability. Swarm intelligence and evolutionary algorithms, such as particle swarm optimization (PSO), or genetic Algorithms (GA), optimize deployment and load balancing, although they suffer from convergence and overhead issues [81,83,97]. Reactive protocols like AODV are favored for their low overhead, but high mobility can lead to frequent route breakage [72]. Hybrid schemes attempt to reconcile the trade-off between proactive and reactive behaviors [98]. Novel concepts like diffusion-based load balancing and joint location and resource optimization (JLRO) for hybrid RF/FSO links reflect the growing complexity and adaptability of network-layer decisions [13,99].

Table 6
Communication protocols reported for UAV emergency networks by predominant OSI layer: functions, advantages, and drawbacks.

OSI layer	Protocol	Function	Advantages	Drawbacks	Refs.
Physical	LTE (E-UTRA radio interface)	High-rate radio access for UAV links (PHY/MAC stack used here by predominant radio function).	High spectral efficiency; mature ecosystem.	Degradation in obstructed/NLoS environments.	[37,54,82]
	Free Space Optics (FSO)	Optical backhaul/access for high-speed links.	Very high bandwidth; immune to RF interference.	Weather-dependent; strict LoS/alignment.	[9,29,69,70]
	Visible Light Communication (VLC)	Optical downlink/uplink using RGB-LEDs between UAVs and ground.	High throughput; RF-immune.	Sensitive to ambient light and pointing errors.	[64]
Data link	IEEE 802.11 (Wi-Fi)	Wireless access/MAC between UAVs and ground stations.	Widely available; easy deployment.	Interference-prone; limited range vs. cellular.	[60,80]
	TDMA	Time-slot allocation to avoid collisions.	Predictable access; reduces collisions.	Requires tight synchronization.	[77]
	EC-CMAC	Energy-aware channel access for UAV networks.	Improves energy efficiency; lowers contention.	Control overhead; complexity under high dynamics.	[71]
	LoRaWAN (LPWAN MAC)	Long-range, low-power MAC/network services for disaster telemetry.	Very low power; extended coverage in sparse deployments.	Very low data rate; duty-cycle limits; unsuited to high throughput.	[70,73]
Network	AODV	On-demand ad-hoc routing for mobile UAV networks.	Low overhead when idle.	Route discovery delay in highly dynamic topologies.	[72]
	Q-learning-based routing	Learns routes under changing conditions.	Adapts to dynamics; potential energy gains.	Computation cost; slow convergence in large graphs.	[46,68,89]
	Delay-Tolerant Networking (DTN)	Store-and-forward for intermittent connectivity.	Robust in disconnected/disrupted networks.	High latency; storage overhead.	[44,57]
	Multipath opportunistic routing	Multi-hop paths leveraging opportunistic forwarding.	Resilient in dynamics; reduces single-point failures.	Computationally demanding in real time.	[57]

Note: When a protocol spans multiple layers, classification follows its predominant function in the reviewed system.

Table 7
Distribution of studies by OSI layer focus in UAV emergency communications (multi-label).

OSI layer	References	%
Physical layer	[9,29,37,54,64,69,70,82]	17.0
Data link layer	[60,71,73,77,80]	10.6
Network layer	[44,46,57,68,72,89]	12.8

Despite their importance, only a few studies address transport or application-layer protocols. Transmission control protocol (TCP) is rarely suitable in UAV networks due to its poor performance over lossy wireless links. User datagram protocol (UDP), combined with error correction at the application layer, is preferred for critical time-sensitive services [100,101]. Application-layer designs often focus on emergency-specific requirements such as traffic prioritization, low-overhead telemetry, or lightweight publish/subscribe schemes for real-time updates. Partial offloading architectures [77] and epidemic routing schemes [44] demonstrate adaptive computation and message delivery in intermittently connected networks. However, only 4% of the studies engage in this layer, highlighting an underexplored area crucial for mission success.

Cross-layer optimization emerges as a vital theme, enabling protocols to coordinate across the OSI stack. For example, MAC decisions can influence routing, and UAV trajectories can affect link availability. Some studies apply reinforcement learning (RL) or deep learning (DL) to jointly adapt modulation, power control, and routing. While promising, these approaches must ensure reliability, interpretability, and real-time performance to be viable in emergencies. The absence of standardized cross-layer frameworks underscores the need for modular, adaptive architectures that integrate physical link quality, MAC scheduling, and network routing. As emergency systems assisted by UAVs evolve, convergence across layers, driven by scenario awareness and AI-based optimization, will be central to ensuring a robust, low-latency, energy-aware communication infrastructure.

4.1.4. Analysis Research Question 2

From these results, it is possible to observe that the protocols used for emergency scenarios are primarily focused on the physical layer, since this layer directly determines how signals are transmitted, received, and propagated between UAVs. The above is crucial in an unstable and infrastructure-deficient environment. However, the performance of physical layer protocols depends on the specific environment. There are many isolated solutions targeting specific layers or issues without a unified architecture. Moreover, many of the proposed protocols are considered in simulation environments without a real-world implementation, which helps to understand their reliability and integration challenges.

4.1.5. Research Question 3

RQ3 (*What UAV-based communication architectures are implemented during emergency scenarios?*) was addressed by reviewing the selected studies and extracting the architectural models they implement. Here, a *communication architecture* denotes the structural organization of the network, its topology, the roles and interconnections of its components (UAVs, satellites, base stations, relays, mobile users), and the coordination mechanisms that govern operation [94,102]. Architectures specify node arrangement and function (e.g., UAVs as aerial base stations, relays, or edge-computing nodes), as well as the hierarchy that regulates traffic flows [33,49,103]. By contrast, *communication systems* refer

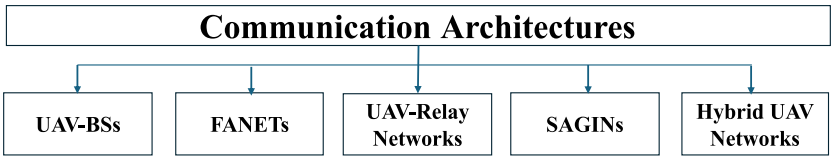


Fig. 6. Families of UAV-based communication architectures in emergencies, highlighting canonical topologies and node roles.

to enabling technologies and protocols (e.g., RF, FSO/VLC, Wi-Fi, LoRaWAN, LTE, 5G/6G) used to exchange data over a given architecture. Although distinct, systems and architectures are interdependent: a single system can support multiple architectural configurations depending on context and operational constraints [6,104,105].

The literature converges on five principal families of UAV-based architectures for emergency communications (Fig. 6): (i) UAV-assisted aerial/mobile base stations (UAV-BS), (ii) FANETs, (iii) UAV relay networks, (iv) SAGINs, and (v) hybrid multi-link UAV networks (e.g., RF/FSO/VLC). To reduce overlap, we apply four operational rules for classification: primary UAV role, topology and hop count, control and coordination mode, and link heterogeneity. The *Hybrid* label is assigned only when the design includes explicit link selection (switching) or joint-use logic.

1. **UAV-Assisted Aerial Base Station (UAV-BS) Networks:** Refers to infrastructure-dependent architectures where UAVs provide direct access to UEs and rely on terrestrial or satellite backhaul. Topology is typically star or point-to-multipoint with minimal inter-UAV coordination. The objective is rapid service restoration or coverage extension for first responders and affected populations [106].
2. **Flying Ad Hoc Networks (FANETs).** Decentralized multi-hop aerial networks in which UAVs act as peers, collaboratively managing routing and relaying without fixed infrastructure. Defining traits are self-organization, distributed control, and adaptability to rapidly evolving disaster conditions [107].
3. **UAV Relay Networks.** Architectures that employ UAVs as intermediate nodes to bridge otherwise disconnected components (users, ground stations, command centers). Unlike FANETs, relay networks here are typically one- to few-hop, planned or supervised, and their primary function is backhaul extension or gap bridging under LoS constraints [108].
4. **Space-Air-Ground Integrated Networks (SAGINs).** These are Multi-layered architectures where UAVs serve as base stations or relays interconnecting satellite, aerial, and terrestrial layers. SAGINs aim for coverage continuity, throughput scaling, and resilience across large or infrastructure-deficient regions, with cross-layer coordination of resources [109].
5. **Hybrid UAV Networks (RF/FSO/VLC/VLC).** Architectures that integrate multiple transmission media (commonly RF and FSO) with *explicit* link-selection/switching or joint-use mechanisms to adapt to environmental and operational conditions [110,111]. Hybrid operation coexists with UAV-BS, Relay, FANET, or SAGIN when such logic forms part of the architectural design.

Table 8 reports the distribution of the selected articles across the five families, and Fig. 7 summarizes the key decision criteria and expected overlaps. Because several studies address more than one family, percentages reflect frequency of mention (multi-label). For narrative synthesis requiring a single label, we adopt the predominant architectural role of the UAV.

As illustrated in Fig. 6, UAV-assisted emergency networks can adopt UAV-BS, FANET, UAV-Relay, SAGIN, or hybrid RF/FSO models. Each offers distinct capabilities suited to specific incident scales and infrastructure states during emergency conditions. Table 9 summarizes advantages, limitations, and future perspectives.

In UAV-BS, UAVs equipped with LTE/5G or Wi-Fi temporarily replace damaged ground sites, enabling rapid coverage restoration and seamless integration with existing cellular cores [54,88,112]. Limitations include endurance, single points of failure, and co-channel interference when multiple UAV-BSs operate nearby [11,113]. Current work addresses these via solar-powered UAVs [52,114], or tethered platforms, advanced mobility-aware deployment strategies, and interference mitigation techniques including coordinated multipoint (CoMP) transmission and 5G beamforming [115,116].

FANETs form fully decentralized, self-organizing networks composed of UAVs that communicate through multi-hop links and distributed routing mechanisms. These networks are particularly well-suited for infrastructure-absent scenarios such as search-and-rescue missions or disaster zones, where rapid deployment and high adaptability are critical [117]. FANETs offer inherent scalability and fault tolerance, allowing the network to reconfigure dynamically in response to node failure or mobility [118]. However, maintaining stable links under high-mobility conditions remains a significant challenge, as frequent topology changes increase routing complexity and packet loss [48,117]. Moreover, decentralized coordination introduces computational and communication overhead, especially in large-scale deployments [119]. Recent advances address these challenges through AI-driven routing algorithms (e.g., RL and fuzzy logic) and the incorporation of edge computing, enabling local decision-making and reduced network latency [77,89,120].

In UAV-relay architectures, drones function as airborne relay nodes to extend connectivity between ground units or to a remote base station. These configurations are typically divided into two modes: single-hop relays, suitable for urban or partially connected areas, and multi-hop relays, designed for remote or obstructed environments where direct communication links are not feasible [87,121]. The benefits of relay-based architectures include cost-effectiveness, since full base station capabilities are not required on each UAV-LoS restoration in obstructed terrains, and reduced end-to-end latency by segmenting long transmission paths into shorter, higher-quality hops [113,116]. However, each additional relay node introduces increased system complexity, signal degradation due to cumulative noise and processing delay, and higher energy consumption since each UAV must remain operational to maintain the link [48,119]. To address these limitations, recent research has proposed adaptive relay positioning strategies using optimization algorithms and AI-based control [12,81], as well as energy-aware scheduling and recharging mechanisms, including UAV rotation and solar-powered endurance extensions, to sustain service continuity in long-duration missions [76,114].

SAGIN combine satellite systems, HAPs, UAVs, and terrestrial infrastructure to form a unified multi-layered network architecture capable of delivering ubiquitous and resilient communication services. SAGIN is particularly valuable in large-scale disaster scenarios where terrestrial infrastructure is damaged or inaccessible, offering redundancy through alternative communication paths and seamless vertical integration across layers [7,11,43,109]. By leveraging the wide coverage of satellites, the flexibility of UAVs, and the capacity of ground networks, SAGIN ensures robust connectivity even in harsh or geographically isolated environments. Recent research has proposed tighter integration with LEO satellite constellations, which offer lower latency and improved throughput compared to traditional geostationary systems [3,74]. Moreover, AI-based resource allocation and topology control are

Table 8
Distribution of research works by UAV communication architecture family (multi-label).

UAV-BSs	FANETs	UAV-Relay networks	SAGINs	Hybrid (RF/FSO)
[6,13,43,54,56,58–60, 62–64,72,76,80,83,84, 87,88]	[6,7,12,46,57,67,71,73, 74,79,81,85,89]	[30,43,44,55,65,66,75, 79,82,86]	[37,46,54,58,61,68,77, 78]	[9,29,69,70]
36.73%	**26.53%**	**20.41%**	**16.33%**	**8.16%**

Note: Percentages reflect frequency of mention across studies and may not sum to 100%.

Table 9
Advantages, drawbacks, and future perspectives of UAV communication architecture families.

Architecture	Advantages	Drawbacks	Future perspectives
UAV-BS	Rapid deployment; direct UE access; scalable coverage [88].	Battery endurance; single-point failures; interference in dense scenes [76].	Extended endurance (e.g., solar); 5G/6G-aware interference mgmt; mmWave/beamforming for higher throughput.
FANET	Resilience via multi-hop; scalability; mobility for SAR [57].	Complex collision avoidance/control; real-time compute load; limited bandwidth under heavy traffic [12].	AI-driven decentralized control; embedded MEC to lower latency and offload processing.
UAV Relay	Range extension/backhaul bridging; improved LoS; fewer BS assets [66].	Relay handovers; power cost of persistent relays; degradation over many hops [30].	Adaptive relay positioning; energy harvesting/recharging for sustained operation.
SAGINs	Extensive coverage; redundancy across layers; capacity via offloading [61].	Cost/complexity; latency (satellite segments); cross-layer resource mgmt [78].	LEO integration to reduce latency; AI-assisted orchestration across layers.
Hybrid (RF/FSO)	High throughput and adaptability; RF interference mitigation; resource efficiency [70].	FSO weather/pointing sensitivity; complexity of real-time switching [29].	Autonomous link-switching; weather-aware proactive reconfiguration.

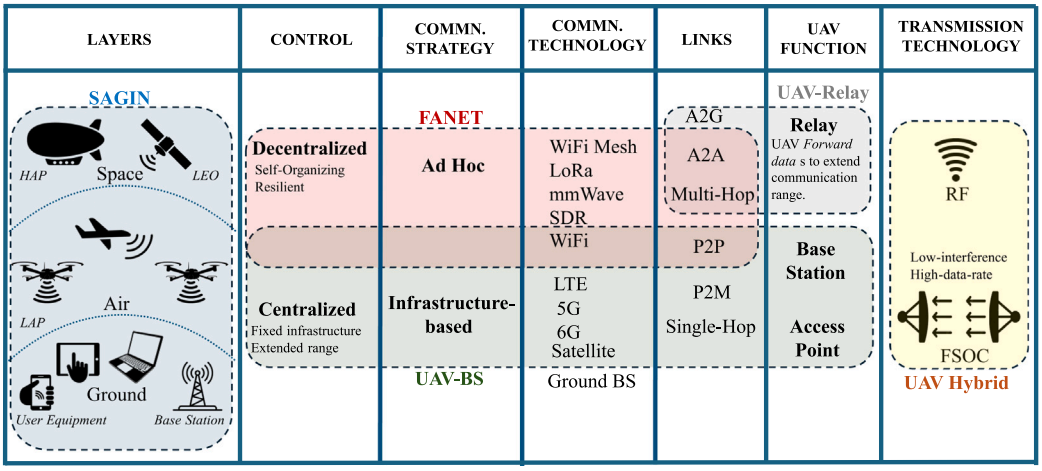


Fig. 7. Decision criteria distinguishing UAV emergency architectures (primary UAV role, hop count/topology, coordination mode, link heterogeneity) and expected overlaps; the hybrid qualifier is cross-cutting.

emerging as key enablers for real-time adaptation to dynamic network conditions and mission priorities [59,89].

Hybrid communication architectures that combine RF technologies (e.g., LTE, Wi-Fi) with FSO are increasingly proposed for UAV-assisted networks to exploit the complementary advantages of both modalities. RF links offer robust connectivity under non-line-of-sight (NLoS) conditions, longer transmission range, and resilience to weather-related degradation, whereas FSO links provide ultra-high data rates, low latency, and immunity to radio frequency interference [9,69,122]. By integrating both technologies, hybrid systems can dynamically switch or aggregate links based on environmental feedback or mission requirements, enhancing overall communication resilience and throughput [70,73]. However, the deployment of FSO on mobile UAV platforms introduces considerable challenges. FSO links are highly sensitive to atmospheric phenomena such as fog, smoke, and rain, and require precise alignment between transceivers to maintain link integrity, especially when both endpoints are in motion [11,64]. These constraints elevate system complexity in terms of both hardware and control algorithms. To address this, current research investigates predictive link adaptation algorithms, multi-beam and beam-steering FSO systems, and atmospheric compensation techniques to sustain FSO performance during dynamic conditions [69,70,74].

4.1.6. Analysis Research Question 3

This survey reveals that UAV-BS architectures dominate (37%), followed by FANETs (27%), UAV-relay (20%), SAGIN (16%), and hybrid (8%). Architecture selection should align with incident scale, infrastructure status, and data requirements. UAV-BS is favored for moderate-scale or partially connected zones; FANETs for autonomous, low-infrastructure scenarios; UAV-relay for range extension; and SAGIN for broad, infrastructure-less areas. Hybrid systems are optimal when high-capacity or interference mitigation is required. However, there is a lack of methodological frameworks for selecting one of these UAV communication architectures. To address this, Fig. 8 proposes a

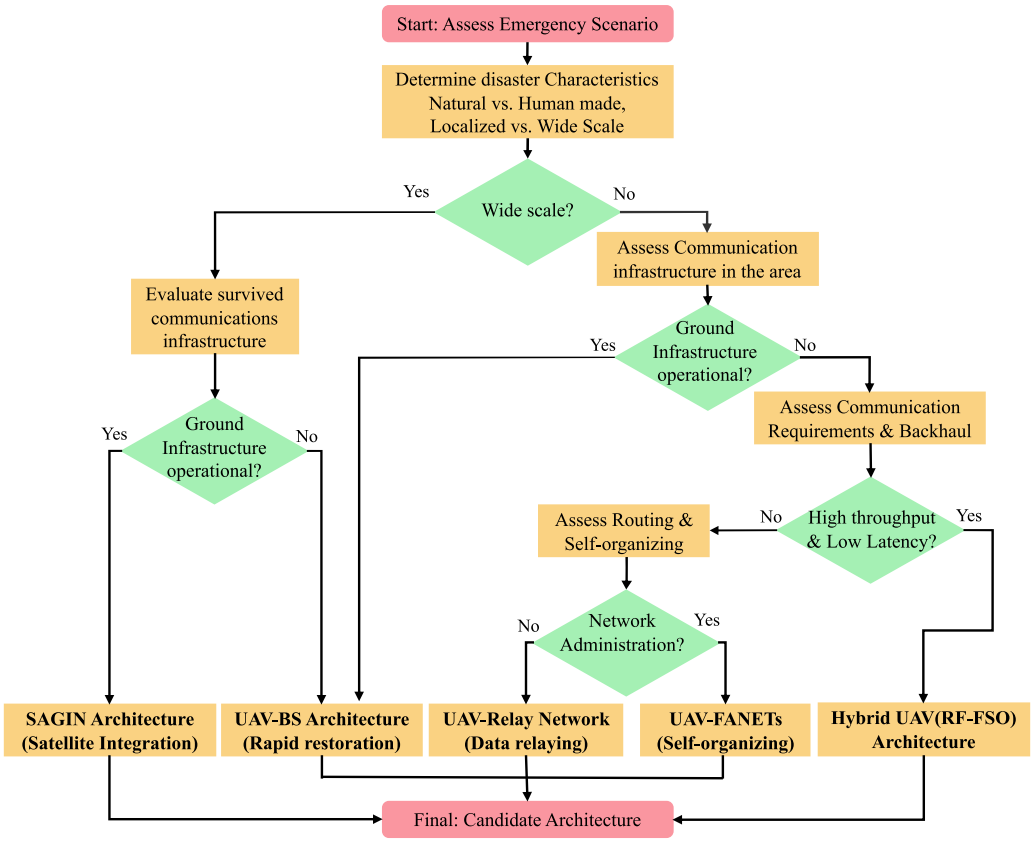


Fig. 8. Architecture selection flowchart for UAV-assisted emergency communications as a function of incident scale, residual infrastructure, latency/throughput targets, and energy budget.

decision-making flowchart to assist in identifying the most appropriate UAV-assisted communication architecture based on the disaster profile.

Recent studies have proposed several innovative enhancements to UAV-assisted emergency communication systems aimed at overcoming energy, latency, and adaptability limitations. One such approach involves tethered UAVs, which are physically connected to ground-based power sources through cables (see Fig. 9), thereby eliminating battery constraints and enabling prolonged airborne operation [123]. This architecture is particularly advantageous in persistent surveillance and prolonged disaster recovery operations where downtime must be minimized.

Another promising direction involves AI-enhanced UAV-BS systems, which leverage ML algorithms to dynamically adjust UAV positioning, transmission power, and routing decisions based on real-time user demand, mobility patterns, and environmental conditions [59,87,89]. These adaptive mechanisms can significantly reduce latency, improve responsiveness, and optimize resource utilization, especially in rapidly changing or resource-constrained disaster zones. RL, particularly deep Q-learning and multi-agent reinforcement learning (MARL), has shown effectiveness in real-time UAV deployment and decision-making, outperforming static or rule-based systems in terms of throughput and coverage quality [7,77,88].

4.2. Optimized communication parameters for UAV-based emergency communications

RQ4 and RQ5 examine (i) the key communication parameters considered in UAV-based emergency communication studies and (ii) the methodologies and algorithms employed to optimize them. We summarize how specific parameters such as transmit power, altitude, and 3D placement are selected or adapted, and we outline the computational techniques proposed to support those choices.

4.2.1. Research Question 4

To address RQ4 (*Which UAV communication parameters can be modified and optimized in emergency contexts?*), we reanalyzed the 47 selected studies and classified them by the parameters they target for optimization. Eight primary parameter families were identified: spatial deployment, link budget and coverage, QoS and traffic performance, MAC/PHY resources, mobility and trajectory, network scale and load, power and endurance, and topology and grouping.

Spatial deployment refers to the placement of UAV platforms in 2D or 3D space, namely altitude and horizontal position, subject to operational and energy constraints. Link budget and coverage includes RF metrics that characterize link quality, such as coverage probability; received signal strength indicator (RSSI, typically ranging from −120 to 0 dBm); signal-to-noise ratio (SNR, in dB); path loss (PL); and effective isotropic radiated power (EIRP). QoS and traffic performance encompasses service-level indicators including end-to-end latency, throughput (data rate), packet delivery ratio (PDR), packet error rate (PER), and hop count in multi-hop settings. MAC/PHY resources denote the allocation and adaptation of communication resources at the medium-access and physical layers (e.g., transmit power, bandwidth, time/frequency slots, and modulation-coding choices) to track time-varying demand. Mobility and trajectory covers flight-path variables for real-time or delay-tolerant operation, with typical goals of reducing latency, energy expenditure, or coverage gaps [124]. Network scale and load captures the relationship between fleet size (e.g., number of UAVs) and user demand, seeking the minimum assets required to serve the largest feasible user set. Power and endurance focuses on energy consumption and battery autonomy for the airframe and payload, with targets such as mission duration and energy per bit. Finally, topology and grouping concerns the network's structural organization (e.g., clustering, role assignment, formation flying) to improve coverage, balance load, or

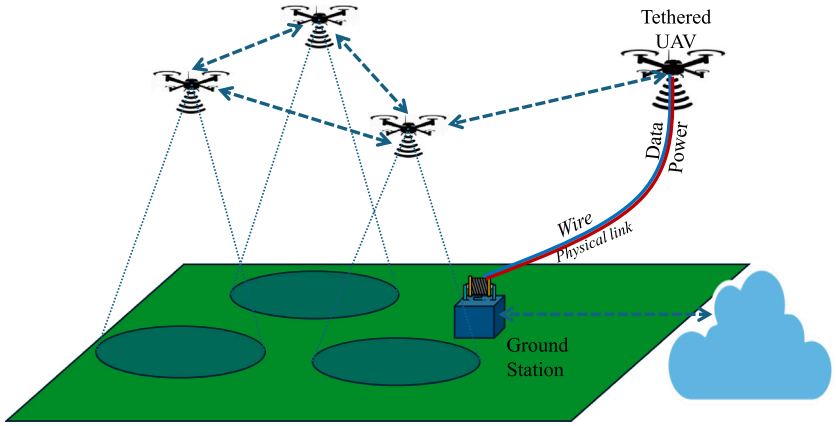


Fig. 9. Tethered UAV concept for persistent emergency operations with continuous ground power and optional data backhaul.

Table 10
Per-study summary of modified parameter families in UAV-assisted emergency communications.

Year	Refs.	Spatial deployment	Link/coverage	QoS/Traffic	MAC/PHY resources	Topology/groupings	Mobility/trajectory	Scale/load	Power/endurance
2014	[79]	✓		✓	✓				
2016	[54]	✓	✓		✓				✓
2018	[84]		✓					✓	✓
	[88]	✓			✓		✓		
2019	[81]	✓		✓	✓	✓			
	[66]	✓	✓	✓			✓		
	[43]	✓		✓	✓		✓		
2020	[9]	✓			✓				✓
	[73]	✓			✓				
	[60]	✓		✓	✓				
	[44]	✓		✓			✓		✓
	[62]	✓	✓		✓		✓		✓
2021	[82]	✓	✓		✓				
	[30]	✓		✓	✓	✓			
	[58]	✓		✓	✓				
	[13]	✓		✓	✓	✓	✓	✓	
	[85]	✓	✓	✓					✓
	[12]	✓	✓					✓	✓
	[72]	✓		✓	✓				✓
	[7]	✓		✓		✓	✓	✓	✓
2022	[65]	✓	✓			✓	✓		
	[46]			✓					
	[67]		✓	✓					
	[29]		✓		✓			✓	
	[86]		✓	✓				✓	
	[76]	✓		✓		✓			
	[37]			✓					
	[64]	✓		✓					
2023	[78]	✓		✓	✓	✓			
	[55]	✓		✓	✓				✓
	[6]	✓		✓	✓	✓			
	[83]	✓	✓	✓	✓				
	[74]	✓			✓		✓		
	[68]	✓		✓				✓	
	[61]	✓	✓				✓		
	[59]	✓			✓		✓		
	[57]	✓	✓	✓			✓		
	[75]	✓		✓	✓				✓
	[70]	✓		✓	✓		✓		
	[69]	✓	✓	✓	✓				✓
	[63]	✓	✓				✓		
	[89]	✓	✓	✓	✓		✓		
2024	[80]	✓		✓	✓				✓
	[56]	✓			✓	✓			✓
	[87]	✓	✓	✓	✓		✓		
	[71]	✓	✓	✓	✓		✓	✓	✓
	[77]	✓	✓	✓	✓				

Note: A checkmark indicates the paper modifies the corresponding parameter family; multiple families may be addressed per study.

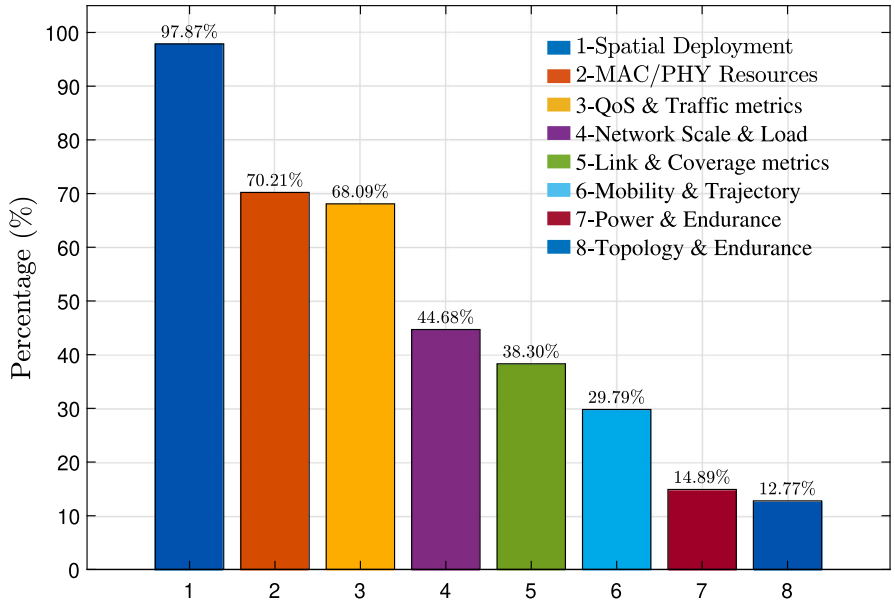


Fig. 10. Share of studies modifying each parameter family to enhance UAV-assisted emergency communications.
Note: Parameter families: spatial deployment; MAC/PHY resources; QoS/traffic performance; scale and load; link budget and area coverage; mobility and trajectory parameters; power and endurance metrics; topology and grouping parameters, (percentages can exceed 100% because many studies target multiple categories).

reduce overhead. Table 10 shows which of the studies focus on each family.

Fig. 10 reports the percentage that spatial deployment is the most frequently modified family (97.9%), followed by MAC/PHY resources (70.21%) and QoS and traffic performance (68%). Network scale and load metrics are about 44.7% of the studies, Link capacity and coverage parameters are 30%, and Mobility and trajectory parameters 29.8%. By contrast, power and endurance (14.9%) and topology and grouping (12.8%) appear less often as primary decision variables.

In addition to categorizing the parameters adjusted in these works, we examined how those adjustments influence key performance outcomes. Fig. 11 provides a high-level overview in which the rows correspond to common performance indicators (network throughput and reliability, RF link quality and coverage, latency, energy efficiency, and number of users served) and the columns correspond to the eight parameter families defined above. This comparative view helps identify which parameter choices are most effective for specific objectives in emergency scenarios. Three consistent associations emerge: (i) adjustments in spatial deployment and mobility and trajectory tend to produce gains in coverage and aggregate throughput; (ii) power and endurance parameters correlate strongly with energy-related key performance indicators (KPIs); and (iii) network scale and load together with topology (Table 11).

Despite these trends, improvements in scalability and multi-user capacity are frequently constrained by interference and backhaul limits. For example, changes to spatial deployment improve signal quality and stability in most cases (reported as 89%) yet have a limited direct effect on user capacity (11%). Similarly, MAC/PHY resource adaptation enhances performance in many studies (66%) but does not by itself overcome spectrum or interference ceilings. Tuning QoS and traffic performance commonly improves coverage (43%), latency (38%), and energy use (32%), although effects on capacity remain modest. Increasing network scale and load can improve redundancy and resilience (observed in 43%) but may exacerbate interference at higher UAV densities. Link budget and coverage adjustments, such as power control and altitude selection, improve RF quality and delay in a comparable fraction of studies (43%) while offering limited scalability benefits. Mobility and trajectory optimization is reported in roughly a third of the corpus, with measurable gains in coverage (26%) and smaller reductions in delay (15%), at the cost of additional computational complexity. Power and endurance appears in fewer works (15%), often focused on power control and lightweight protocols, where battery constraints limit absolute gains. Topology and grouping is the least explored family (13%); it provides localized benefits in energy consumption and delay but has so far shown limited impact on system-level scalability.

4.2.2. Analysis Research Question 4

Taken together, these findings indicate that per-family adjustments, although effective for coverage, latency, and energy, do not, on their own, resolve the fundamental constraints of capacity and scale under emergency conditions. The evidence points to the need for integrated, context-aware policies that jointly coordinate placement, routing, and resource control across the eight families, with explicit handling of interference and backhaul bottlenecks. Such joint designs are more likely to translate the gains observed in controlled evaluations into robust, large-scale performance in the field.

Across the reviewed studies, spatial deployment is the most frequently adjusted parameter family, followed by MAC/PHY resource allocation and QoS/traffic performance, while power/endurance and topology/grouping are examined less often as primary design variables. Three regularities are evident: (i) modifying spatial placement and trajectories tends to increase coverage and aggregate throughput; (ii) altering power and endurance parameters primarily improves energy-related KPIs; and (iii) decisions on scale, load, and topology most strongly affect the number of users served and overall robustness. However, gains in scalability and multiuser capacity are often limited by interference, backhaul constraints, and air and ground spectrum contention, and many results are obtained only in simulation. These observations point to the need for integrated, context-aware policies based on joint optimization of placement, routing, MAC/PHY control, and energy-aware trajectory planning, with validation in heterogeneous, large-scale deployments under realistic resource constraints.

Designing UAV networks involves navigating competing demands; therefore, this review identifies the next key trade-offs, as follows (See Fig. 12):

- **Coverage vs. Energy Consumption:** Maximizing coverage drains battery life. Evolutionary algorithms and RL approaches balance these by adjusting UAV roles and mobility patterns dynamically.

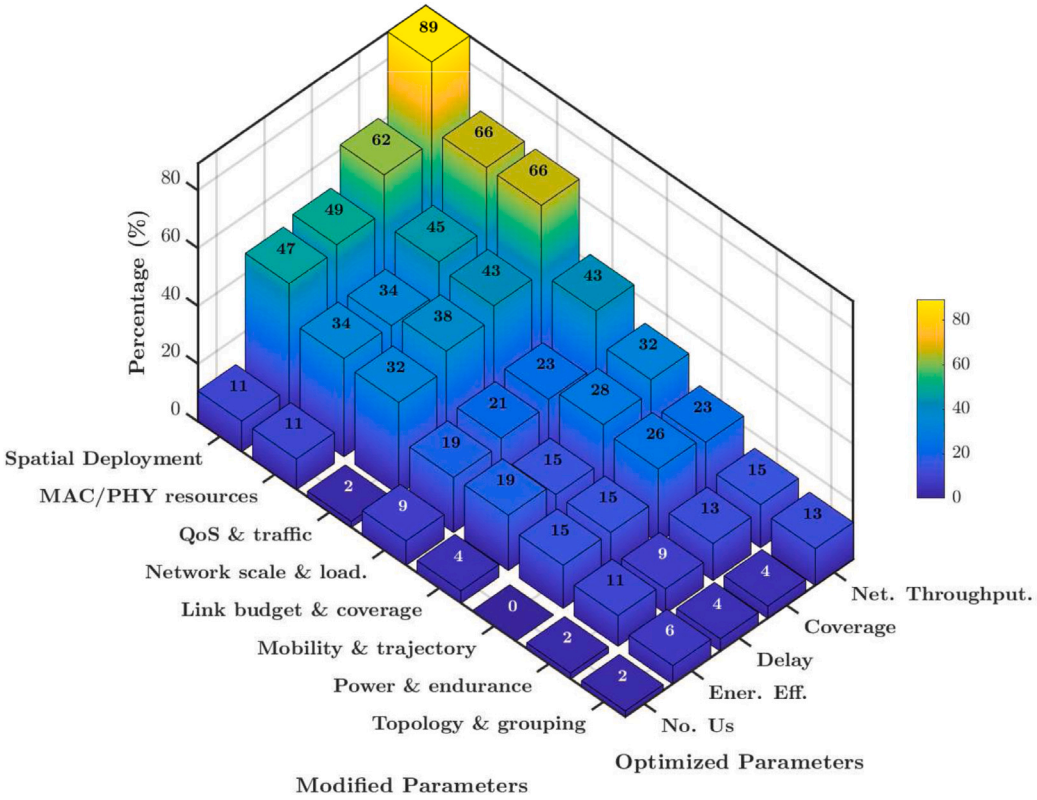


Fig. 11. Reported performance gains by outcome (rows) versus modified parameter families (columns) across the reviewed studies.
Note: The Outcomes include network throughput/reliability, RF link quality and coverage, latency, energy efficiency, and number of users served.

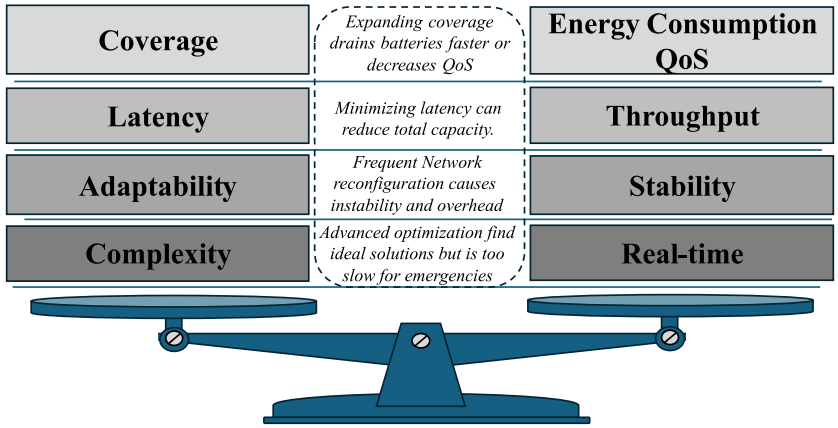


Fig. 12. Key trade-offs regarding UAV-assisted Emergency Communication Systems identified in the literature.

- **Latency vs. Throughput:** High data rates may introduce delay; low latency paths might compromise capacity. Multi-objective optimization and heuristic strategies are used to strike this balance.
- **Adaptability vs. Stability:** Frequent reconfiguration ensures responsiveness but risks instability. Threshold-based updates or predictive control offer compromise solutions.
- **Coverage vs. QoS:** Wide area service may reduce link quality. Adaptive UAV scheduling and phased deployment address this by adjusting density and mission focus over time.
- **Computational Complexity vs. Real-Time Feasibility:** While exact methods offer high accuracy, heuristics and hybrid approaches (e.g., quick heuristic first, refined by RL) support real-time operations. Edge computing can help offload intensive tasks when connectivity allows.

4.2.3. Research Question 5

To address RQ5 (*What optimization techniques and methodologies are applied to enhance UAV communication performance during emergencies?*), an analysis was conducted to identify the algorithms, optimization techniques, and mathematical tools employed to solve challenges associated with UAV-based communication systems in emergency scenarios. The analysis considered the technical contributions, methodologies, optimization techniques, and specific algorithms or methods described in the literature. Additionally, the investigation examined optimized parameters, interference and channel management strategies, application areas, and the objectives outlined in each study.

The identified algorithms were classified into five distinct categories, each representing the primary optimization objective addressed by the respective studies. These categories were defined based on their

Table 11
Key performance Indicators (KPIs) and metrics classification and descriptions.

KPI Category.	Description.	Metrics.	Refs.
Coverage & Connectivity.	Effectiveness in restoring or establishing reliable wireless coverage after infrastructure failure in post-disaster areas. Focuses on geographic reach, number of users served, and signal quality.	- Users covered (%) - Coverage area (km^2) - RSSI, SNR	[6,12,37,43,54,56,61,62,76,83,85–88]
Throughput & Data rate performance	Capacity and efficiency of data transmission and ability to handle significant amounts of data quickly for real-time information exchange during emergencies.	- Aggregate throughput - User data rate - BER, PDR	[7,29,59,64,68–71,78–80,82,89].
Latency & Delay minimization.	Responsiveness for real-time communications, essential for timely decision-making, effective management, quicker information dissemination and lower response times during emergency situations.	- End-to-end delay - Task delay (Edge) - MAC delay.	[7,46,57,58,71–73,75–77,79,83,89,109].
Reliability, stability & Robustness.	Capability to maintain system stability and consistent performance under adverse conditions (weather, interference, mobility) and ensure uninterrupted communication in disaster scenarios.	- Link stability - Route stability - PDR, BER.	[29,37,44,57,60,61,64,67,69,79,80,89].
Energy efficiency & UAV endurance.	Effective energy resources management and UAV operational lifetime. Strategies designed to maximize battery life and minimize power consumption.	- Energy consumption - Battery lifetime - Efficiency per bit.	[7,30,58,71,75–78,83,84].
Fairness & QoS.	Equitable network resource allocation, traffic prioritization, and QoS differentiation so that no group or type of traffic is systematically disadvantaged.	- Fairness index - QoS guarantees - Priority handling	[13,59,65,70,71,74,75,79,87,88].
Computational & Algorithmic efficiency.	Computational overhead and effectiveness of algorithms used for routing, positioning, and clustering in UAV networks, ensuring rapid and accurate deployment decisions.	- Convergence speed - Computational complexity - Localization accuracy	[7,12,56,59,66,72,74,78,79,81,85,87,89].
Channel & Propagation characteristics.	Radio propagation analysis and characterization for accurate channel modeling and improving communication performance under realistic conditions.	- Delay spread - Temporal correlation - Channel accuracy	[44,54,60,61,69,80,82].
Deployment & Operational flexibility.	Evaluates how quickly and effectively UAV systems can be deployed and adapted to varying disaster scenarios, including dynamic environments, heterogeneity, and mobility constraints.	- Deployment time - Number of UAVs - System integration	[6,13,37,43,56,60,65,73,76,79,83,88].
Data processing & Edge computing.	Local (at the network edge) data processing and offload tasks capabilities for real-time decisions to reduce latency and bandwidth usage.	- Edge latency - Offloading ratio - Server utilization	[59,60,72–77,79,82,83].

focus on spatial, energy, spectral, or task-based optimizations, carefully considering the optimization techniques, algorithms employed, modified and optimized parameters, and the characteristics, applications, advantages, and limitations of these methodologies. Efforts were made to minimize overlap between categories to reduce redundancy while preserving critical information. This classification emphasizes the core optimization goals and techniques relevant to UAV-based communication networks, providing a comprehensive understanding of the methodologies used. Each category reflects the unique features of the optimization approaches and underscores their applicability to real-world network operations. These categories are Coverage and positioning optimization, Energy efficiency and power management, Interference and channel management, Routing and path planning, Data offloading and Edge Computing, Quality of Service and Task Allocation, and Signal Integrity and Channel Modeling, in which the articles can be classified, as shown in Table 12. In addition, Fig. 13 shows the percentage of research work on each observed approach. Finally, Tables 13 and 14 show a detailed description of the studies according to the categorization presented in the last table.

The first categorization is coverage and positioning optimization, which analyzes the types of optimization algorithms and techniques that take advantage of UAVs' flexibility to change position in 3D space, controlled by human operators or even by algorithms, giving them a certain degree of movement autonomy. The second category is Energy Efficiency, which is related in specific points to the first category but focuses on minimizing energy consumption for UAV motion or communication systems to extend battery life while maintaining the effectiveness of communication by keeping communication requirements or Quality of Service, taking into consideration modifying parameters like transmission power, relay selection, sleep mode duration, adaptive power control for finding UAVs routes or hovering positions based on energy efficiency criteria. Articles classified into these categories are presented in Table 13.

The routing and path planning category is another central recurring approach, which optimizes data routing or UAV path planning to improve communication efficiency and reduce latency, throughput, or other network parameters. Data offloading transfers data processing or communication tasks from one system element to another to optimize performance, reduce congestion, reduce latency, or save energy consumption. It ensures efficient utilization of scarce resources and maintains QoS or network requirements in UAV-assisted communication, particularly in emergency scenarios, supporting real-time applications. Signal integrity is another perspective, which improves signal strength and reliability, mainly in difficult environments with high interference, obstructions, or insufficient communication infrastructure. However, articles classified in the category of interference and channel management focus on reducing signal interference, particularly in highly dense UAV or hybrid networks, for multi-UAV swarm-based deployments to enhance or recover communications performance. Finally, Quality of Service and task allocation are studied in a limited number of references (2) to ensure a specific quality of service required

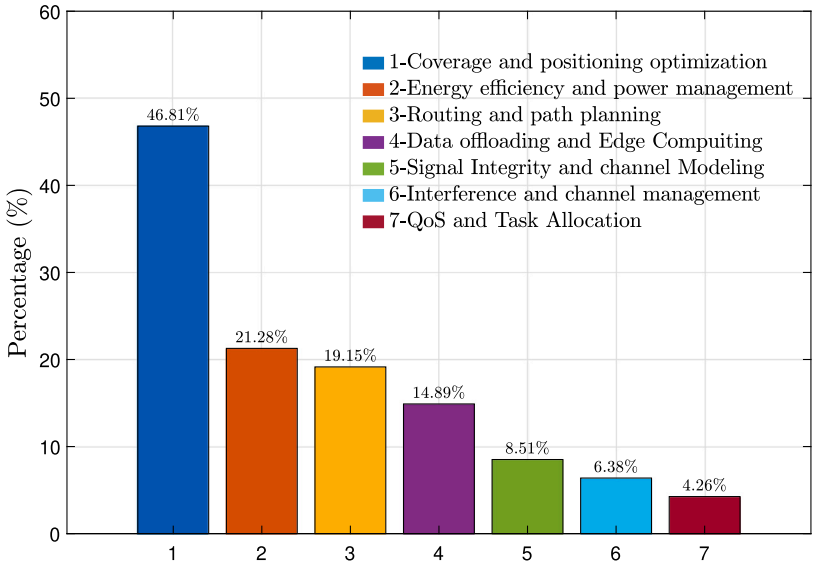


Fig. 13. Distribution of optimization methodology categories applied in UAV emergency communications.
Note: Percentages can exceed 100% because many studies target multiple categories.

Table 12
Optimization tasks.

Category	References	%
Coverage and positioning optimization	[6,9,12,13,29,43,44,54,56,58,60,62,63,70,79,80,82,83,85,86,88,125]	46.8%
Energy efficiency and power management	[29,30,55,56,59,71,78,81,83,84]	21.3%
Routing and path planning	[7,46,55,57,67,68,79,87,89]	19.1%
Data offloading and edge computing	[37,72–77]	14.9%
Signal integrity and channel modeling	[61,64,66,69]	8.5%
Interference and channel management	[6,54,65]	6.4%
QoS and task allocation	[9,13]	4.3%

Note: Percentages can exceed 100% because many studies target multiple categories.

or expected for a communication system, mainly to maintain the performance needed for critical tasks or search and rescue missions (as shown in Table 14). All results are discussed later.

Based on Table 12 and Fig. 13, it is possible to observe that the majority of UAV-assisted emergency communication studies focus on optimizing coverage and positioning. However, critical aspects such as interference management and channel modeling remain underexplored. Beyond positioning, the literature addresses energy efficiency, routing, task offloading, signal integrity, and QoS, employing a variety of algorithmic approaches (Table 14).

Coverage and Positioning (46.8%): Techniques include k-means clustering, PSO, GAs, and RL (e.g., Q-learning, deep Q-Networks (DQN)). These aim to maximize area coverage and minimize energy consumption or overlap, often in terrain-sensitive or user-dense environments.

Energy Efficiency and power management (21.3%): Optimization methods include adaptive power control, energy-aware MAC protocols, and EH. Hybrid strategies such as Dinkelbach's algorithm with DRL jointly optimize trajectory and resource allocation [59].

Routing and Trajectory Planning (19.1%): Approaches include geographic routing, multi-hop protocols, and Q-learning-enhanced path planning. Objectives center on minimizing latency and maximizing throughput through intelligent waypoint selection and multi-objective criteria.

Edge Computing and Offloading: Roughly 15% of works use edge-based or federated learning methods to reduce latency and bandwidth use. Techniques involve task scheduling via Markov chains [73], swap matching [74], and integration with IoT and MEC frameworks [72,75–77].

Signal Integrity and Channel Modeling (8.5%): Includes neural network-based channel estimation, PSO-optimized SDM, and empirical models for rugged terrains. Goals are reliable high-quality links, especially in complex environments.

Interference and Channel Management (6.4%): Though rarely addressed, some studies apply game theory, GA with REB, or joint altitude/resource allocation (JARA) to manage spectrum access and reduce interference in dense deployments.

QoS and Task Allocation (4.3%): Scheduling and prioritization methods aim to meet low-latency, high-reliability requirements, particularly in search-and-rescue and control communications.

4.2.4. Analysis Research Question 5

Despite their potential, most optimization strategies are validated through simulations or analytical modeling. Real-world testing remains rare. Studies like [61,64] demonstrate feasibility through field measurements, but broader deployment trials are needed.

In summary, a variety of optimization methods have been explored to manage these trade-offs. Early works often used classical optimization or heuristics, while more recent studies increasingly leverage artificial intelligence techniques (ML, DRL) for adaptive optimization. An interesting observation is that relatively few approaches have been validated outside of simulations. While algorithms like RL show promise in theory, their performance in real-world conditions (with unpredictable factors and failures) remains largely unverified. This connects to the research gap of experimental validation discussed later. Nonetheless, the trend is toward multi-objective optimization frameworks that can juggle several metrics (coverage, latency, energy, throughput) simultaneously, reflecting the multifaceted needs of ECSs. The use of hybrid

Table 13
Optimization methodologies-Part I: coverage and positioning; energy efficiency and power management.

Category	Optimization methodology	Refs.
Coverage and positioning optimization	Genetic Algorithm (GA) for optimizing UAV placement, Range Expansion Bias (REB) to balance traffic loads.	[54]
	RESCUE algorithm: QoS-aware drone base station placement and mobile user association strategy used to determine the 3-D locations of DBSs, the mobile user association, and the bandwidth allocation in access links to maximize the number of served mobile users.	[9]
	UAV deployment algorithm for positioning UAVs to maintain network coverage, optimizing resource utilization and network performance.	[79]
	Dense Multi-Agent Reinforcement Learning (DMARL) algorithm for multi-UAV deployment and dynamic trajectory control.	[80]
	Cluster-based communication algorithm for optimizing coverage and dynamic routing adjustments based on UAV positions.	[6]
	Two-Layer Maximum Coverage Deployment Problem (2L-MCDP-C&U): Variable Neighborhood Search integrating k-means (VNS-K).	[56]
	Swarm Intelligence-based Localization (SIL) and Clustering (SIC) for energy-efficient UAV positioning using PSO.	[125]
	Iterative alternating algorithm that minimizes transmission delay by simultaneously optimizing UAV positioning, bandwidth allocation, and user power control for THz communication.	[82]
	Two-stage hierarchical approach: SAIBFGS for small areas and LAESGA for large-scale energy-efficient UAV deployments.	[83]
	Diffusion-based UAV deployment integrating virtual force fields and successive convex optimization for fairness in multi-UAV networks.	[13]
	Hybrid metaheuristic algorithms for optimal 3D UAV placement: HWWO-HSA for medium/large scale, HGA-SA for small scale.	[12]
	RESTORE algorithm with NLoS-FSOC: Optimizing UAV positioning and service coverage using TLA, PLA, and PMA policies.	[29]
	Combine 3D location, power allocation, and bandwidth for throughput maximization and QoS satisfaction in UAV-based networks.	[58]
	Matching Game Algorithm for dynamic UE-DSC association and optimal resource allocation in UAV networks.	[60]
	Metaheuristic multi-objective algorithms (NSGA-II, MOPSO, SPEA2) for optimal UAV placement in aerial mesh networks.	[85]
	Voronoi-based flight method for delay-tolerant aerial networks incorporating user density functions and dynamic destination selection.	[86]
	Adaptive flight schemes using DTN, Epidemic Routing, AODV, and waypoint/rebounding flight for UAV positioning in AWRNs.	[44]
	Distributed Q-learning-based reinforcement learning for UAV positioning and DSC deployment to maximize coverage.	[88]
	Joint Location and Resource Optimization (JLRO): Optimize UAV 3D positioning, power, and bandwidth allocation for FSO and RF links.	[70]
	Joint trajectory and communication scheduling optimization to maximize coverage and throughput while ensuring QoS.	[43]
	UAV flight formation and conformal antenna design optimizing coverage, reliability, and flexibility using CMA.	[62]
	k-means++ clustering and antenna adjustment for multi-UAV positioning and performance optimization.	[63]
Energy efficiency and power management	CHERA framework: Combines K-means++ clustering, Lagrangian-dual method, and Dinkelbach algorithm for BS sleeping and power allocation in a Self-Organized Network (SON) framework.	[78]
	Multi-objective path planning: Uses clustering, adaptive resource allocation, and dynamic positioning with route adaptation for energy-efficient hybrid communication networks.	[55]
	Clustering algorithm: Optimal selection of cluster heads (CHs), EH using SWIPT, and D2D communication strategies for sustainable and energy-efficient systems.	[30]
	Variable Neighborhood Search (VNS-K): Integrates altitude adjustment strategies to reduce UAV energy consumption while maintaining coverage and connectivity in a two-layer Capsule Airport (CA) system.	[56]
	Swarm Intelligence-Based optimization: Applies SIL (Localization) and SIC (Clustering) using PSO to form scalable, energy-efficient UAV clusters and select CHs.	[81]

(*continued on next page*)

Table 13 (*continued*).

Category	Optimization methodology	Refs.
	Adaptive launch power and hover height adjustments: Incorporates environmental factors and user dispersion into UAV deployment to improve energy efficiency.	[83]
	RESTORE algorithm: Integrates NLoS-FSOC, TDMA scheduling, and dynamic altitude adjustments to optimize energy efficiency.	[29]
	Deep Reinforcement Learning (DRL-EC3): Optimizes UAV movement for energy efficiency and ensures effective and fair communication coverage.	[84]
	DDinkelbach's method with DDQN: Combines resource allocation and trajectory optimization for superior energy efficiency in UAV-assisted 5G networks.	[59]
	Energy Consumption and Channel Gain Cooperative MAC (EC-CMAC): Balances energy consumption and communication performance by integrating channel-aware relay selection and adaptive transmission strategies.	[71]

Note: Entries summarize the core technique, decision variables, and intended effect.

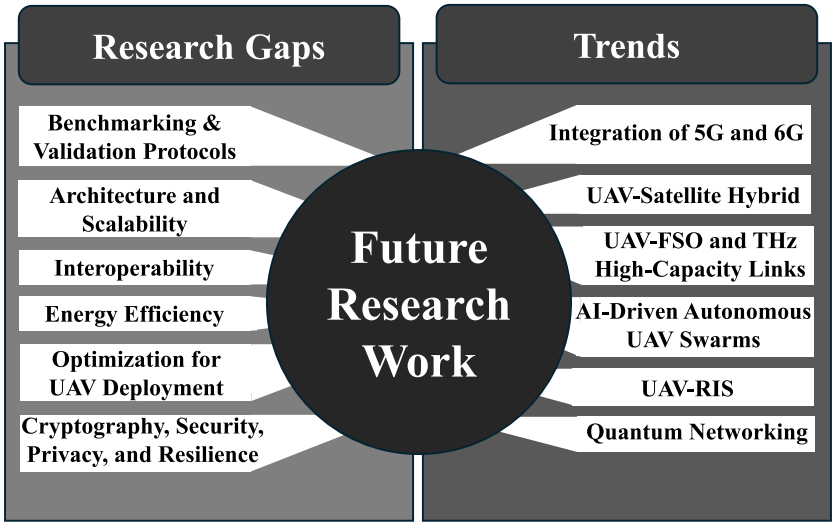


Fig. 14. Research gaps and emerging trends in UAV-assisted emergency communication systems reported in the literature.

optimization methods (e.g., combining an initial heuristic with subsequent ML adjustments) appears to be a practical path to achieve robust solutions that meet the demands of disaster scenarios in real time.

5. Knowledge gaps

In UAV-assisted emergency communication architectures, clear applicability criteria must be defined according to the type and scale of the incident. While SAGINs provide wide-area coverage via multilayer integration, their deployment entails substantial infrastructure and operational costs, which may limit suitability for sudden or highly localized disasters. In such cases, simpler designs (e.g., UAV relay networks) are often preferable when demand is moderate or when parts of the terrestrial infrastructure remain operational.

Although numerous optimization methods have been proposed to configure UAV communication parameters during emergencies, most evaluations rely on simulations under idealized assumptions that do not capture uncertainty and operational complexity in real scenarios. Physical deployment and field testing remain limited. A core obstacle is the absence of standardized, scenario-driven test protocols and datasets that emulate disaster dynamics, hindering reproducibility, fair comparison, and confidence in experimental results.

Addressing these gaps requires coordinated, interdisciplinary work. Researchers, industry, regulators, and first responders should co-develop technical and operational guidelines enabling realistic, reproducible, and mission-driven validation of UAV communication systems in emergency contexts [52]. The key research gaps and emerging trends derived from this survey are outlined below and summarized in Fig. 14.

5.1. Research gaps

The synthesis of RQ1–RQ5 suggests that progress in UAV–assisted emergency communications is constrained less by a lack of algorithmic novelty than by shortcomings in validation practices, architectural generality, cross-technology operability, energy-aware design, trustworthy artificial intelligence, and cyber-resilience. Across the corpus, most reported gains are established in simulation under assumptions that do not reflect the volatility of disasters, the scarcity of backhaul and power, or the mixture of traffic classes that first responders face. Architectural proposals are seldom tied to explicit incident profiles or to reproducible selection rules that relate residual infrastructure, spatial scale, and QoS objectives to a concrete topology. Interoperability across multiple radio access technologies (RATs) is acknowledged as necessary, yet procedures for seamless vertical handover and cross-layer coordination are rarely demonstrated beyond small examples. Energy remains a binding constraint, and only a minority of studies treat communication control and power management as a single coupled problem with measurable sensitivity to environmental and mission variables. Artificial intelligence methods deliver large simulated gains but lack hardware-in-the-loop (HIL) and field validation that would show robustness under sensor noise, delayed rewards, and strict size-weight-power budgets. Finally, the treatment of security, privacy, and continuity under failure or attack lags behind protocol and optimization advances. The following subsubsections delineate the specific gaps and the methodological depth needed to close them.

Table 14
Optimization methodologies-Part II: routing and mobility; data offloading and edge computing; signal and interference; QoS and task allocation.

Category	Optimization methodology	Refs.
Routing and path planning	Geographic routing protocol: Determines local routes for data flow based on UAV and node positions; incorporates Bellman-Ford variant for global optimal route computation.	[79]
	Multi-layered strategy: Integrates waypoint planning, trajectory optimization, and network configuration optimization for efficient data collection and transfer.	[55]
	Multi-Hop Opportunistic 3D Routing (MO3DR) Algorithm: Manages dynamic topology, inter-UAV collision avoidance, and dynamic coverage in routing.	[57]
	Position-aware routing protocols: Employs Q-learning and multi-objective optimization for routing in highly dynamic and resource-constrained FANETs.	[46]
	Hypergraph matching algorithm: Optimizes multipath routing in UAV swarm networks, addressing dynamic path planning and routing challenges.	[67]
	Q-Learning framework: Balances energy consumption, throughput, and reliability in a hierarchical aerial-ground network through dynamic learning rate adjustments.	[68]
	Heterogeneous Coordinated QMIX (HC-QMIX): Multi-agent deep reinforcement learning (MADRL) for decentralized routing and UAV trajectory optimization combined with user association.	[87]
	Improved PSO (IPSO) algorithm: Combines Variable Neighborhood Search (VNS) for optimal cluster head selection and routing path optimization.	[7]
	Q-Learning-Based Routing (QRIFC): Uses 2-hop packet transmission and Adaptive Flocking Control (AFCA) to coordinate UAV swarm with dynamic topology and energy efficiency.	[89]
Data offloading and edge computing	Integration of drone edge intelligence and IoT-based wearable devices for real-time data processing, low latency, and efficient resource utilization during emergency operations.	[72]
	Markov chain-based task management and a three-tier network model leveraging UAV-MEC and LoRaWAN for balancing real-time data offloading and computational efficiency.	[73]
	Swap matching-based task scheduling algorithm and SCA-Based Multi-UAV Deployment for efficient data offloading and edge computing.	[74]
	Heuristic task offloading algorithms (Aggressive-Wait, Aggressive Tx-Order, Adaptive) and Quasi-Optimal Offloading (Q-OPT) for task prioritization, dynamic resource allocation, and real-time edge processing.	[75]
	Intelligent UAV edge computing and ad hoc networking to optimize data offloading and resource management.	[76]
	Air-Ground Integrated Mobile Edge Network (AWMEN) combining Deep Reinforcement Learning (DRL) with TD3-BC-R and Federated Learning (FL) for optimal data offloading.	[77]
	Ubiquitous Sensor Networks (USNs) integrating edge computing, NVIS, Drone-Based Wireless Mesh Networks, and IoT for efficient data offloading.	[37]
Signal integrity and channel modeling	Neural network-based signal modeling: Three-layer neural network for signal integrity modeling, with linear regression for comparison.	[66]
	Optimization for channel modeling: Combines improved Particle Swarm Optimization (PSO) and Space Division Multiplexing (SDM) to enhance channel efficiency.	[69]
	Mobility models for signal reception: Markov c[66]ins for UAV trajectory modeling and Smooth-turn (ST) mobility model for ground signal optimization.	[61]
	Comprehensive signal processing: Includes YOLOv3 for object detection, 8B/10B encoding, MIMO equalization, and OOK modulation to improve signal integrity.	[64]
Interference and channel management	Genetic Algorithm (GA): Optimizes UAV placement and incorporates Range Expansion Bias (REB) to enhance network performance while balancing interference.	[54]
	Joint Altitude and Resource Allocation (JARA) algorithm: Adjusts UAV altitude and allocates resources to mitigate interference and manage channels effectively.	[65]
	Cluster-based communication algorithm: Employs adaptive backoff mechanisms and dynamic resource allocation to minimize channel contention and interference while ensuring reliable communication.	[6]
QoS and task allocation	Algorithm: Optimizes drone base station (DBS) placement and mobile user association to ensure QoS, maximize the number of served mobile users, and optimize bandwidth allocation.	[9]
	Diffusion UAV deployment algorithm: Incorporates virtual force fields and a distributed diffusion strategy to optimize UE allocation and ensure data rate fairness in multi-UAV networks.	[13]

Note: Methodologies are grouped by primary task; studies may span multiple tasks.

5.1.1. Benchmarking and validation protocols (cross-cutting)

Based on the results addressed in RQ1 and RQ2, the evidence remains predominantly simulation-based and often depends on idealized propagation, benign mobility, and stationary traffic. As a result, the results are not directly comparable between studies, and external validity is limited. The literature includes valuable physical-layer demonstrations, for example, outdoor optical camera communication between UAVs, yet these isolated pilots do not constitute a maturation pipeline from simulation to HIL and then to controlled field trials [64]. Existing demands for coordinated, mission-driven validation highlight the need for standard artifacts and shared practices [52], but widely accepted benchmarks are not yet available. A credible benchmarking suite should define incident classes with parameterized disruption patterns, publish mobility traces for hover, patrol, and swarm manoeuvres, and provide failure injections for backhaul loss, power scarcity, and sensor faults. It should also standardize KPIs and reporting, including spatial user coverage, percentile end-to-end latency, packet delivery ratio, aggregate throughput, route and link stability over time, control overhead, and energy per bit. Crucially, protocols and algorithms should be evaluated under identical channel models, terrain layouts, and workload mixes, with openly accessible configurations and seeds to enable exact replication. Without this scaffolding, it is difficult

to translate the many positive simulation outcomes into dependable operational guidance.

5.1.2. Network architecture and scalability

From RQs 3 and 5, it is observable that Architectural contributions often optimize single-UAV or small-fleet deployments without addressing the reconfiguration requirements of real incidents. SAGINs provide broad coverage and redundancy, yet introduce orchestration complexity and cost that may be unsuitable for sudden or highly localized events. Relay oriented aerial networks are effective when part of the terrestrial infrastructure remains operational or when demand is moderate, whereas FANETs and UAV-BS overlays target larger or rapidly evolving footprints. What is still missing is a principled mapping from incident profile to architectural family, together with transition rules that govern migration as conditions change. Such a mapping should consider incident scale, residual infrastructure, latency and throughput targets for distinct traffic classes, and the available energy budget. Software-defined networking (SDN) and network function virtualization (NFV) are repeatedly proposed to manage heterogeneous topologies and to enable flexible resource allocation, but empirical evidence on controller placement under mobility, failover times, multi-tier orchestration overheads, and their impact on quality of service remains scarce [37]. Moving beyond frequency counts of architectural families requires publication of explicit selection policies accompanied by scalability studies that expose stability regions and degradation modes under load and failure.

A related limitation concerns interoperability at the system boundary. Cognitive radio strategies and multi-band platforms are natural remedies for heterogeneous conditions, yet the literature rarely reports reproducible procedures for cross-technology handovers between UAV and satellite, UAV and LTE or 5G, or UAV and LoRa, with consistent metrics for application-level loss, handover rate, and spectral efficiency [95,126,127]. In addition, cross-layer couplings are acknowledged but remain weakly standardized. Trajectory and altitude decisions that modify contention at the medium access layer and route stability at the network layer should be formalized within architectural descriptions and evaluated under shared scenarios to enable portability and fair comparison across studies.

5.1.3. Communication technologies and interoperability

The survey confirms in RQ1 and RQ2 active research at the physical, data link, and network layers, yet practical interoperation across access technologies remains limited. Many implementations are anchored to a single RAT such as LTE, 5G, WiFi, or LoRa, which reduces adaptability when channel conditions, user density, or backhaul options change. In this context, multi-RAT operation denotes the capability of a platform to sense, select, and when appropriate, combine multiple RATs, for example LTE or 5G with WiFi or LoRa and satellite links, while preserving end-to-end service continuity. It is included here because emergency response environments are heterogeneous and time varying, so continuity depends on seamless vertical handover and on exploiting complementary links without application level disruption. Cognitive radio mechanisms and multi-band front ends are promising directions, but the literature rarely reports reproducible handover procedures for UAV to satellite, UAV to LTE or 5G, or UAV to LoRa, with consistent metrics for outage, handover rate, buffering, and spectral efficiency [95,126,127]. Cross layer couplings also remain underspecified. Trajectory and altitude decisions that modify contention at the medium access layer and route stability at the network layer should be codified as part of the design and evaluated under shared scenarios with common datasets and KPIs. Without such standardization, claims of interoperability are difficult to compare, reproduce, or adopt in practice.

5.1.4. Energy efficiency and UAV endurance

Energy constraints cut across architectures and protocol stacks were analyzed in RQ4 and RQ5. From the above, While trajectory, It was seen, that altitude, and placement optimizations are common; comparatively few studies implement joint control of communication and energy that adapts transmit power, duty cycles, link duration, and task offloading to battery state, mission urgency, and channel variability [7,59]. Recent joint formulations show that coupling these degrees of freedom can improve both QoS and endurance, yet most evidence is still simulation-based and lacks sensitivity analyses that explore adverse weather, payload changes, and traffic surges [77,87]. Hardware approaches such as solar-assisted platforms and tethered UAVs extend endurance in specific envelopes, but quantitative characterizations of their limits and tradeoffs are sparse. These include the influence of irradiance, temperature, aerodynamic penalties from panels, maximum tether length, mechanical reliability, and the properties of the tether-based backhaul [76,123]. Future studies should report energy per delivered bit, mission duration at specified traffic loads, and recovery behavior under energy scarcity, all under common scenarios, to support design choices that generalize beyond single case studies.

5.1.5. Artificial intelligence in real-world optimization

Artificial intelligence consistently improves trajectory planning, scheduling, and resource allocation in simulation studies, with RL, DQL, and MARL showing clear benefits for coverage, latency, and load balancing [7,89] (see RQ1 and RQ5 results). The main open questions concern robustness and feasibility in real environments. Convergence properties under sensor noise and delayed rewards are not fully characterized, and stability during exploration remains uncertain for safety-critical missions where policy updates can perturb flight control or link maintenance. In addition, many proposals do not quantify computational and energy budgets on constrained avionics, which limits transfer from laboratory settings to deployable systems [59,87].

Federated learning is attractive for local adaptation, with lower backhaul and higher privacy. However, its performance in airborne networks with intermittent connectivity and non-identically distributed data has not been rigorously established. Open issues include client selection under sporadic links, the resilience of aggregation to lagged data, bias and drift due to heterogeneous data distributions, and the communication cost of repeated rounds with tight mission timelines [77,128–130]. Aside from notable physical layer pilots demonstrating the feasibility of airborne optical links [64], there is still no standardized maturation path connecting simulation to HIL and micro-scale controlled field trials.

A credible path forward is a validation pipeline that specifies, for each learning task, the admissible state and action spaces, the safe exploration policy, and the compute and memory ceilings that reflect realistic airframes and payloads. Such a pipeline should define environmental disturbances and failure injections, for example, sensor bias, GPS outages, packet loss bursts, and energy throttling, together with mission-level KPIs for coverage, delay, packet delivery, and autonomy loss. It should also prescribe fallback behaviors under sensing or control degradation and require public implementations of training and evaluation code to enable independent reproduction. Establishing this pipeline would allow the community to compare algorithms on common ground and to demonstrate that the gains observed in simulation translate into robust, efficient, and certifiable behavior in the field.

5.1.6. Security, privacy, and network resilience

Security and service continuity have not advanced at the same pace as protocol design and performance optimization. Open wireless operation to jamming, spoofing, and eavesdropping [90,131]. Proposed countermeasures span decentralized authentication with blockchain, anomaly detection aided by ML, lightweight cryptography tailored to small aerial platforms, hardware roots of trust through physical unclonable functions, and preparation for post quantum cryptography [37,75,

128,132–134]. What remains insufficient is systematic evaluation that quantifies the practical overheads and protection levels under realistic conditions. Few studies report added latency and energy attributable to security functions, PDR under targeted interference or deception, false positive and false negative rates for anomaly detection, or key establishment time measured on airborne hardware rather than desktop proxies.

Resilience mechanisms such as multipath routing, failover redundancy, and distributed control require stress testing under controlled combinations of mobility, interference, and targeted attacks [76]. Future studies should adopt shared scenarios and uniform reporting templates that specify at a minimum the traffic mix, explicit attack scripts with intensity and timing, radio and mobility traces, and a complete set of KPIs with confidence intervals and effect sizes.

Cryptographic evaluations should disclose computational cost per operation, memory footprint, and end-to-end blockchain-based throughput measured on representative flight controllers rather than desktop surrogates. For blockchain-based authentication, reports should include consensus time, message overhead, resource utilization, and robustness to network partition and membership dynamics. For anomaly detection, authors should provide calibration procedures, receiver operating characteristic curves, class imbalance handling, and performance degradation when telemetry is delayed, missing, or corrupted. Only with comparable and reproducible evidence along these dimensions will it be possible to assess and certify robustness for deployment in safety-critical emergencies.

5.2. Emerging trends

The current state of research reveals persistent technical gaps alongside a set of promising directions that could materially enhance UAV-assisted communications for emergency response. The items below synthesize trends that recur across the reviewed literature and that are directly aligned with the operational constraints identified in this study.

5.2.1. Integration of 5G and 6G networks

5G systems with URLLC communication support time-critical coordination and control [11,135]. Looking toward 6G designs, advances under active study include THz bands [136], RISs [137], joint communication and sensing also referred to as ISAC [138,139], and AI-assisted orchestration across the stack. These capabilities are expected to improve reliability, throughput, and adaptability under surge demand in emergencies, provided that their control and measurement interfaces are exposed for robust use in rapidly changing environments.

5.2.2. UAV–satellite hybrid communication systems

LEO constellations offer backhaul when terrestrial infrastructure is impaired. Realization in emergencies requires routing and handover procedures that remain stable under variable delay, intermittent visibility, and rapid motion [140,141]. Effective hybrids combine the spatial flexibility of UAV platforms with the ubiquity of space segments, including explicit policies for path selection, buffering, and congestion control when switching between air, space, and ground links.

5.2.3. High-capacity FSO and THz links

FSO links can deliver high data rates with low latency but degrade under fog, rain, or dust; RF and FSO hybrids mitigate weather sensitivity by switching or aggregating links [69,70]. THz communications in the 0.1 to 10 THz range promise extreme rates and fine spatial resolution for sensing [142–145]. Key challenges include beam alignment under mobility, transceiver power consumption, and limited aerial endurance. Cooperative relaying and the use of intelligent surfaces have been proposed to improve robustness [146,147]. Progress will depend on broader field validation, aerial channel models specific to these bands, and harmonized safety assessments [142–144].

5.2.4. Autonomous UAV swarms enabled by AI

Multi-agent control and swarm intelligence enable decentralized coordination that adapts to user density, obstructions, and mission priorities, thereby improving resilience and scalability during dynamic incidents [80,81,89]. Reported methods optimize topology, routing, and resource allocation in real time. To reach operational maturity, these approaches should expose stability margins, convergence behavior, and compute and energy budgets on representative avionics, and they should include fallback policies when sensing is degraded.

5.2.5. UAV-Assisted Reconfigurable Intelligent Surfaces (RIS)

Intelligent surfaces can shape propagation to create effective LoS paths, suppress interference, and increase spectral efficiency with modest power budgets. Mounting such surfaces on UAVs combines mobility with programmable channels to sustain low latency and high reliability in disrupted areas. Studies report gains in coverage, outage reduction, and efficiency, and explore simultaneously transmitting and reflecting surfaces for broader angular support as well as swarm-assisted aerial surfaces for larger apertures and robustness [148–151]. Methodical evaluation under motion, blockage, and wind-induced vibration remains a priority.

5.2.6. Quantum networking

Quantum networks leverage qubits (superposition and entanglement) to enable quantum key distribution (QKD) and distributed quantum information processing [152,153]. UAVs can serve as agile line-of-sight (LoS) mobile nodes, connecting ground units, extending coverage, and rapidly providing secure links in critical scenarios. [154,155]. Persistent challenges include decoherence and channel impairments in free-space links (pointing, tracking, turbulence), as well as size, weight, and power limitations, and cooling and power requirements for onboard sources and detectors. Ongoing work on miniaturized photonic hardware, quantum/AI co-design for adaptive link control, and quantum-secure registration advances future, resilient command and control systems.

6. Conclusions and future work

This study surveys the existing literature regarding UAV deployment in emergency communication scenarios, comprehensively covering application contexts, communication protocols, architectures, and optimization techniques. The analysis reveals that UAV-based communication solutions predominantly address large-scale disaster scenarios, with relatively limited attention given to integrating mobile edge computing and real-time communication capabilities, both of which are crucial for rapid response in dynamic and unpredictable disaster environments.

In terms of communication protocols, most reviewed systems are focused on the physical, data link, and network layers. Furthermore, commonly adopted architectures include UAV-BSs, FANETs, and relay-based configurations. Predominant optimization targets identified were coverage, throughput, and latency, often pursued through heuristic methods such as PSO, GA, and artificial intelligence-driven approaches (i.e., RL and Deep RL), focusing on optimal UAV placement and resource management.

However, despite the wide range of communication protocols and architectures, and optimization techniques used to set the parameters for deploying UAVs to enable communication systems during emergencies, we notice a lack in real-world implementations of the proposed methods. Hence, there is an opportunity to propose and generate guidelines that allow the real-world implementation and validation of UAV communication systems that also enable reproducible and reliable results.

CRediT authorship contribution statement

Isaac López-Villegas: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Erick Axel Martínez-Rios:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Javier Izquierdo-Reyes:** Writing – review & editing, Supervision, Conceptualization. **Rogelio Bustamante-Bello:** Writing – review & editing, Supervision, Conceptualization. **Francisco Falcone:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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